

Evaluating Hedge Fund Performance with Prudent Investors*

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Abstract

We propose a new performance evaluation method which allows for investors with skewness preference, i.e., prudent investors. Applied to hedge funds, our method yields different conclusions compared to standard tests, especially in the recent subsample, for younger funds, and compared to mutual funds. We also derive a performance measure, prudent alpha, which generalizes simple alphas to prudent investors. For hedge funds, the difference between simple and prudent alphas is economically significant and positively related with withdrawal restrictions but unrelated to managerial incentives. For simulated option strategies, prudent alpha is better able to assess performance than existing option-based factor models.

JEL Classification: G10, G11.

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The empirical literature on hedge funds has long emphasized that hedge funds have highly non-linear, option-like returns which may arise through the use of dynamic trading strategies, derivatives, short-selling, and leverage.¹ Option-like returns make it challenging to assess whether or not hedge funds improve the investment opportunity set of investors. In particular, the usual performance evaluation approach of regressing fund returns on benchmark asset returns and advising investors to buy the funds which have a positive alpha relies on the assumption that the first two moments accurately summarize the return distribution for investors. It is well known that this assumption is not satisfied if returns have option-like distributions.

This paper develops a general framework to assess whether or not assets with option-like returns improve the investment opportunity set of investors. Our framework generalizes the standard performance evaluation approach based on regressions and explicitly takes into account investors' preferences beyond the first two moments of returns. We use our framework to better understand the costs and benefits of higher moments in hedge fund returns compared to mutual fund returns and to shed new light on the debate of the usefulness of hedge funds from a portfolio perspective. We also show in a simulated economy that our approach is better able to assess the true alpha of option trading strategies than the option-based factor models which are widely used in hedge fund research.

Evaluating investments from a portfolio perspective is in line with Ferson and Lin [2014] who argue that alpha should be defined using the “client’s utility function” to be meaningful. In this vein, we extend the mean-variance spanning and intersection approach of Huberman and Kandel [1987]² to investors who care for higher moments of returns and, more precisely, to prudent investors.

Prudence refers to a positive third derivative of the utility function [Kimball, 1990] and measures

¹See, e.g., Fung and Hsieh [1997, 1999, 2001], Agarwal and Naik [2004], Stulz [2007], Jurek and Stafford [2015].

²The mean-variance spanning approach has been widely used in the literature to assess international diversification benefits [see, e.g., Bekaert and Urias, 1996, Errunza et al., 1999, DeRoos et al., 2001] and the regression-based tests of mean-variance spanning and intersection are related to tests of mean-variance efficiency, performance measurement and portfolio choice [DeRoos and Nijman, 2001].

the preference for skewness and the aversion to downside risk of investors.³ In terms of portfolio choice, prudent investors solve a trade-off between a higher expected return, a lower variance and more right skewness. As a theoretical contribution, we derive the conditions under which new assets improve the investment opportunity set of prudent investors.

In our empirical analysis, we study the costs and benefits of option-like characteristics in returns with a large sample of live and dead hedge funds and mutual funds from Morningstar in the period from 1994 to 2014. We thereby consider investors who currently invest in stocks and bonds. We obtain different results with our method compared to the standard spanning tests: among the hedge funds with less (more) pronounced option like characteristics and compared to mean-variance investors, less than half (more than twice) as many funds improve the investment opportunity set of prudent investors. These differences are also more pronounced for younger funds and in the recent subperiod. More generally, prudent investors are able to improve their investment opportunity set with more than 30% of the hedge funds but with only less than 10% of the mutual funds. These appealing hedge funds tend to be smaller, have higher performance fees and higher minimum investment requirements. In terms of investment category, hedge funds in the event, global derivatives, multistrategy and relative value categories are more likely to improve the investment opportunity set, and funds of funds are less likely to improve the investment opportunity set.

Next we examine the economic magnitude in terms of utility cost of option-like features in fund returns by comparing simple alphas to alphas adjusted for skewness preference, i.e., prudent alphas. The prudent alpha is obtained from the intersection conditions of our framework and is the return an asset delivers in excess of the required return by the investor. Mathematically, a prudent alpha is the sum of the simple alpha and an adjustment proportional to co-skewness. Prudent investors

³To be precise, the statement "all prudent investors prefer payoff A to payoff B " is only sufficient for the statements "payoff A is more skewed to the right than payoff B " and "payoff A has lower downside semi-variance than payoff B " [Menezes et al., 1980]. This is similar to risk aversion, i.e., a negative second derivative of the utility function, in the sense that the statement "all risk averse investors prefer payoff A to payoff B " is only sufficient for the statement "payoff A has a lower variance than payoff B " [Rothschild and Stiglitz, 1970].

are willing to forgo almost 2.5% return per year on average to hold a hedge fund in the high co-skewness decile rather than a fund in the low co-skewness decile. For mutual funds simple and prudent alphas differ less and prudent investors are willing to forgo less than 1% return per year on average in the cross-section of funds. We also analyze the relationship between co-skewness and hedge fund characteristics and find that hedge funds with longer lock-up and advance notice periods have more negative coskewness. This result is consistent with the theoretical model and the empirical results in Hombert and Thesmar [2014] who suggest that funds with contractual impediments to withdrawals optimally take more downside risk.

To validate our approach of ignoring moments above the third order, we also derive alphas adjusted for moments up to kurtosis and alphas adjusted for all higher moments and compare them to the prudent alphas. We find that ignoring the fourth and higher moments has very little impact. The difference between prudent alphas and alphas adjusted for all moments is mostly below 0.10% per year and prudent alphas therefore seem to provide an accurate summary of the option-like characteristics in hedge fund returns.

As a final test, we apply our approach to the returns of option strategies such as buy-write and protective puts in a simulated economy with a log-normally distributed market return. In such an economy, a positive (negative) simple alpha and negative (positive) co-skewness is achieved with strategies selling (buying) put or call options. We verify that in our approach the alphas of buy-write and protective put strategies are close to zero. In addition, we find that previously proposed factor models which include option returns or polynomials of the market return as factors are unable to assess that the alpha of these strategies is zero. Notice that buy-write strategies have the same payoffs as the put-writing strategies considered in Jurek and Stafford [2015] and these findings therefore suggest that our approach can assess whether or not a hedge fund alpha is merely a compensation for bearing market downside risk.

The remainder of the paper is organized as follows. The next section situates our paper in the literature of performance evaluation and hedge funds. Section 2 contains the theoretical framework and discusses the tests. The empirical evaluation of hedge funds performance is in Section 3 and Section 4 concludes. Appendix A relates our mean-variance-skewness framework to the expected utility framework, and the proofs are in Appendix B.

1 Related literature

Our research is related to the literature on hedge fund performance and risk evaluation. This literature finds that hedge funds are different from mutual funds because additional risk-factors are needed to explain their returns⁴ and their risk exposures show strong time-variation [Bollen and Whaley, 2009, Patton and Ramadorai, 2013]. Even after controlling for additional risk factors, previous research finds that a subset of hedge funds delivers alpha [Kosowski et al., 2007, Buraschi et al., 2014a]. This research builds on the evidence that higher-order risk and tail risk help explain the cross-section of hedge fund returns [Agarwal et al., 2009, Kelly and Jiang, 2012] which suggests that traditional performance metrics may not accurately measure the “alpha” of hedge funds.

The most widely used method to assess hedge fund performance is a linear factor model with the factors of Fung and Hsieh [2001] or Agarwal and Naik [2004]. Motivated by Glosten and Jagannathan [1994], these factors include returns on option based strategies to capture the option-like characteristics of hedge fund returns. However, we show in Section 3.6 that our utility-based approach is better able to assess the true alpha of option strategies than option based factor models which is in line with the results in Leland [1999] and Ingersoll et al. [2007]⁵.

⁴See Fung and Hsieh [1997, 2001], Agarwal and Naik [2004], Bali et al. [2012], Buraschi et al. [2014b] among others.

⁵Ingersoll et al. [2007] propose to rank performance of funds based on the certainty equivalent return calculated with a constant relative risk aversion (CRRA) utility function. Leland [1999] instead generalizes Jensen’s alpha for a CRRA utility function to get a measure which ensures that additional investments improve the overall marginal utility of investors and not just the marginal utility based on the first two moments.

Our approach is close to Leland [1999] and Korteweg and Nagel [2016]⁶ in the sense that our tests assess the (marginal) benefits of investing in hedge funds given a set of existing assets or an existing portfolio. However, we focus on the first three moments which is motivated by several factors. First, a constantly growing empirical, experimental and theoretical literature argues that investors care for skewness⁷. In addition, skewness preference is an important feature of positive theories of investor's choice like cumulative prospect theory [Barberis and Huang, 2008, Ebert and Strack, 2015] and the optimal expectations theory [Brunnermeier et al., 2007, Jouini et al., 2014]. Second, our empirical analysis shows that extending our framework to the next moment, kurtosis, yields an additional kurtosis adjustment which is not economically significant. More importantly, the empirical analysis shows that the extension to skewness captures almost all the difference between mean-variance utility and specific utility functions which account for preferences over all higher moments. Finally, focusing on the mean-variance-skewness framework is more general in the sense that it may approximate also other utility functions and enables us to make statements for all levels of risk aversion and prudence.

Our approach is also related to Mencia and Sentana [2009] who propose a parametric test for mean-variance-skewness spanning when returns follow a multivariate location-scale mixture of normal distributions. Their fully parametric framework differs in important respects from our work because under their distributional assumptions the mean-variance-skewness frontier can be generated by only three funds, which is not the case in general, and hence their spanning conditions are different from the conditions derived in this paper and do not nest the Huberman and Kandel [1987] spanning and intersection tests as a special case.

⁶Korteweg and Nagel [2016] propose the venture capital performance metric "generalized public market equivalent" which takes the multiperiod nature of venture capital payoffs into account and is based on an exponential stochastic discount factor. The single factor and single period version of their method is equivalent to the CRR approach of Leland [1999].

⁷See Arditti [1967], Rubinstein [1973], Kraus and Litzenberger [1976], Horvath and Scott [1980], Harvey and Siddique [2000], Jondeau and Rockinger [2006], Mitton and Vorkink [2007], Guidolin and Timmermann [2008], Martellini and Ziemann [2010] among others; and Ebert [2015] and the references cited therein for experimental evidence.

Existing research on hedge funds from a portfolio perspective finds that hedge funds do not integrate well in a portfolio of stocks and bonds because although they improve the mean-variance trade-off, they do so at the expense of lower skewness [Amin and Kat, 2003]. Our paper provides formal tests to analyze the cost of skewness for investors and finds that the mean-variance benefits usually outweigh the lower skewness and that about 30% of the hedge funds significantly improve the mean-variance-skewness investment opportunity set of stock and bond investors. More recently, Bali et al. [2013] analyze the attractiveness of hedge funds relative to stocks and bonds using almost first- and second-order stochastic dominance and find that some hedge fund styles outperform stocks and bonds. Our approach is conceptually different because first- and second-order stochastic dominance make no assumption on whether or not investors have a preference for skewness whereas we assume that - in line with empirical and experimental evidence - investors do have a preference for skewness. In addition, our tests assess the marginal benefits of hedge funds whereas the stochastic dominance tests implicitly assume that the whole portfolio is invested in hedge funds.

2 Evaluating investments with option-like returns

In this section we describe our performance evaluation approach in a nutshell and then discuss how our conditions are derived from a portfolio choice problem involving mean-variance-skewness utility.

2.1 The approach in a nutshell

Let the vectors of net returns on the k benchmark and n test assets be denoted by $\mathbf{r}_{t+1}^x$ and $\mathbf{r}_{t+1}^y$, respectively. Mean-variance spanning and intersection tests involve the following regression

$$\mathbf{r}_{t+1}^y = \mathbf{a} + B\mathbf{r}_{t+1}^x + \boldsymbol{\epsilon}_{t+1}, \quad (1)$$

where $\boldsymbol{\epsilon}_{t+1}$ is the n -dimensional vector of residuals, $\boldsymbol{a}$ is the n -dimensional vector of intercepts and B is the $n \times k$ -dimensional matrix of slope coefficients. Mean-variance intersection implies that $\boldsymbol{a} + \eta(\boldsymbol{B}\mathbf{1}_n - \mathbf{1}_k) = \mathbf{0}$ for a given value of η and mean-variance spanning implies that $\boldsymbol{a} = \mathbf{0}$ and $\boldsymbol{B}\mathbf{1}_n - \mathbf{1}_k = \mathbf{0}$ [Huberman and Kandel, 1987]. If a risk-free asset is available and in a regression with excess returns over the risk-free rate, mean-variance spanning implies that the intercept is zero. This result provides the rationale to advise investors to buy funds or stocks which have a positive alpha in a linear factor model. For benchmark and test assets with non-normally distributed returns and investors with preferences over higher moments, this result no longer applies. The theoretical contribution of this section is to show that by imposing additional conditions on the $n \times k^2$ -dimensional co-skewness matrix of the residual $\boldsymbol{\epsilon}_{t+1}$ with the benchmark assets, i.e.,

$$\boldsymbol{S}_{\boldsymbol{\epsilon}xx} = \begin{bmatrix} \boldsymbol{S}_{\boldsymbol{\epsilon}xx_1} & \cdots & \boldsymbol{S}_{\boldsymbol{\epsilon}xx_k} \end{bmatrix}, \quad (2)$$

with

$$\boldsymbol{S}_{\boldsymbol{\epsilon}xx_1} = \begin{bmatrix} E[\epsilon_{y_1} \bar{r}_{x_1} \bar{r}_{x_1}] & \cdots & E[\epsilon_{y_1} \bar{r}_{x_k} \bar{r}_{x_1}] \\ \vdots & \ddots & \vdots \\ E[\epsilon_{y_n} \bar{r}_{x_1} \bar{r}_{x_1}] & \cdots & E[\epsilon_{y_n} \bar{r}_{x_k} \bar{r}_{x_1}] \end{bmatrix},$$

we get the conditions for mean-variance-skewness intersection and spanning. For our spanning tests, we calculate the elements of $\boldsymbol{S}_{\boldsymbol{\epsilon}xx}$ with regressions and test the restrictions with Wald tests⁸.

2.2 The theory

Consider the portfolio choice problem of an investor who derives utility from the first three moments of his portfolio returns. The investor can either invest his portfolio in k assets with net returns $\boldsymbol{r}^x$, the benchmark assets, or in a larger universe of $k + n$ assets which consists of n additional assets

⁸The construction of the Wald tests is explained in detail in Karehnke [2014].

which have net returns $\mathbf{r}^y$, the test assets. If the optimal portfolio of the investor is the same with the benchmark assets only and with the benchmark and test assets, the mean-variance-skewness frontier of $\mathbf{r}^x$ and $(\mathbf{r}^x, \mathbf{r}^y)$ intersect [Huberman and Kandel, 1987]. If the optimal portfolio of $\mathbf{r}^x$ and $(\mathbf{r}^x, \mathbf{r}^y)$ is the same for any mean-variance-skewness investor (i.e., for any preference for mean and skewness and any aversion to variance), the benchmark assets are said to span the test assets. In the following, we develop the concepts of intersection and spanning for mean-variance-skewness investors formally.

2.2.1 Spanning and intersection with only risky assets

Let the $k + n$ vector of net returns be denoted by $\mathbf{r}^\top \equiv [\mathbf{r}^{x^\top} \mathbf{r}^{y^\top}]$ and let the vector of expected returns and the matrix of covariances be denoted by $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$, respectively. Bold letters denote vectors or matrices throughout the paper and, if it is not specified otherwise, vectors and matrices have the dimension $(k + n) \times 1$ and $(k + n) \times (k + n)$, respectively. In addition, we sometimes use the subscripts $\mathbf{x}$ and $\mathbf{y}$ to refer to the respective moments of benchmark and test assets, e.g., $\boldsymbol{\mu}_y$ is the vector of the expected returns of the n test assets and $\boldsymbol{\Sigma}_{yx}$ is the $n \times x$ -matrix of the covariances between test and benchmark assets. The $(k + n) \times (k + n)^2$ matrix of co-skewnesses⁹ is given by

$$\begin{aligned} \mathbf{S} &= E \left(\bar{\mathbf{r}} \bar{\mathbf{r}}^\top \otimes \bar{\mathbf{r}}^\top \right), \\ &= \left(\begin{bmatrix} E(\bar{r}_1 \bar{r}_1 \bar{r}_1) & \cdots & E(\bar{r}_1 \bar{r}_1 \bar{r}_{k+n}) \\ \vdots & \ddots & \vdots \\ E(\bar{r}_{k+n} \bar{r}_1 \bar{r}_1) & \cdots & E(\bar{r}_{k+n} \bar{r}_1 \bar{r}_{k+n}) \end{bmatrix} \cdots \begin{bmatrix} E(\bar{r}_1 \bar{r}_{k+n} \bar{r}_1) & \cdots & E(\bar{r}_1 \bar{r}_{k+n} \bar{r}_{k+n}) \\ \vdots & \ddots & \vdots \\ E(\bar{r}_{k+n} \bar{r}_{k+n} \bar{r}_1) & \cdots & E(\bar{r}_{k+n} \bar{r}_{k+n} \bar{r}_{k+n}) \end{bmatrix} \right), \end{aligned}$$

where $\bar{\mathbf{r}}$ are the demeaned returns and $\otimes$ is the Kronecker product.

⁹In statistics, the term skewness is used to refer to the third standardized moment (i.e., the third moment divided by the cube of the standard deviation). Here, we refer to skewness as the third unstandardized moment which is in line with the portfolio choice literature on skewness.

We consider an investor who likes the mean and skewness of his portfolio returns and dislikes the variance of his portfolio returns. The investor chooses his portfolio $\mathbf{w}$ in the $k + n$ assets to maximize his mean-variance-skewness utility

$$\begin{aligned} \max_{\mathbf{w}} \quad & \mathbf{w}^\top \boldsymbol{\mu} - \frac{1}{2} \gamma_1 \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w} + \frac{1}{3} \gamma_2 \mathbf{w}^\top \mathbf{S}(\mathbf{w} \otimes \mathbf{w}), \\ \text{subject to} \quad & \mathbf{w}^\top \mathbf{1} = 1, \end{aligned} \tag{3}$$

where γ_1 and γ_2 are two positive scalars which measure respectively the aversion to variance and preference for skewness (relative to the preference for the mean). In an expected utility framework, γ_1 can be interpreted as the coefficient of risk aversion and γ_2 as one half of the product of the coefficient of risk aversion and prudence or equivalently as one half of the coefficient of downside risk aversion of Crainich and Eeckhoudt [2008]. Appendix A shows how to obtain this interpretation from the first-order condition of an expected utility maximizer and a second-order Taylor approximation of his marginal utility around expected wealth. The appendix also discusses possible values of γ_1 and γ_2 for popular utility functions.

Throughout the paper, we assume that the first-order conditions of the investors are necessary and sufficient, which is the case either if the second-order condition of (3) holds¹⁰, or if we consider the first-order conditions of (3) to be a second-order approximation of the first-order condition of an investor with a concave utility function¹¹. By solving the portfolio choice problem for a specific investor, i.e., a given pair of (γ_1, γ_2) , and imposing the condition that he optimally invests only in the set of benchmark assets, we get the condition for mean-variance-skewness intersection in the next proposition.

¹⁰The second-order conditions require that $-\gamma_1 \boldsymbol{\Sigma} + 2\gamma_2 \mathbf{S}(\mathbf{w} \otimes \mathbf{I})$, where $\mathbf{I}$ is a $k + n \times k + n$ identity matrix, is negative semidefinite for all $\mathbf{w}$. This assumption is a necessary working assumption but may be restrictive when the set of assets allows to form portfolios with very high skewness relative to variance and investors have very large γ_2 relative to γ_1 . We have checked that the second-order conditions are satisfied for the intersection tests in the empirical section.

¹¹This is illustrated in Appendix A.

Proposition 1 *The mean-variance-skewness frontier of $\mathbf{r}^x$ intersects the mean-variance-skewness frontier of $(\mathbf{r}^x, \mathbf{r}^y)$, if*

$$\boldsymbol{\mu}_y - \boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\boldsymbol{\mu}_x + \eta (\boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\mathbf{1}_k - \mathbf{1}_n) + \gamma_2 \{ \mathbf{S}_{yxx} - \boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\mathbf{S}_{xxx} \} (\mathbf{w}_x^* \otimes \mathbf{w}_x^*) = \mathbf{0}, \quad (4)$$

holds for a particular pair of preference parameters (γ_1, γ_2) , which is equivalent to a pair of Lagrange multiplier η and γ_2 , and a corresponding $\mathbf{w}_x^$.*

Proof. See Appendix B. ■

Using (4) and noting that mean-variance-skewness spanning means that any mean-variance-skewness investor only holds the benchmark assets, we obtain the conditions for spanning in the next proposition.

Proposition 2 *The mean-variance-skewness frontier of $\mathbf{r}^x$ spans the mean-variance-skewness frontier of $(\mathbf{r}^x, \mathbf{r}^y)$, if*

$$\boldsymbol{\mu}_y - \boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\boldsymbol{\mu}_x = \mathbf{0}_n, \quad (5)$$

$$\boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\mathbf{1}_k - \mathbf{1}_n = \mathbf{0}_n, \quad (6)$$

$$\mathbf{S}_{yxx} - \boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\mathbf{S}_{xxx} = \mathbf{0}_{n \times k^2}, \quad (7)$$

where the equalities apply element-wise.

Proof. See Appendix B. ■

Notice that our conditions for mean-variance-skewness spanning nest the conditions for mean-variance spanning as a special case. Indeed, we get the conditions for mean-variance spanning, i.e., (5) and (6), if we set γ_2 in (4) to zero. In addition, to see that $\mathbf{S}_{\epsilon xx}$ in (2) contains the restriction in (7) observe that $\mathbf{S}_{\epsilon xx}$ can be rewritten to $E(\epsilon \bar{\mathbf{r}}_x^\top \bar{\mathbf{r}}_x^\top) = \mathbf{S}_{yxx} - \boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\mathbf{S}_{xxx}$.

Extensions of mean-variance intersection and spanning tests to take into account short-sales constraints and transaction costs developed by DeRoos et al. [2001] can be adapted to the mean-variance-skewness case. In the following, we present the extension to short-sales constraints. The spanning restrictions (5) to (7) then have to hold for each subset of the benchmark assets on which the short-sales constraints are simultaneously not binding. Let these subsets be denoted by $\mathbf{x}^j$, for $j = 1, 2, \dots, M$. Using these notations, we can state the following proposition.

Proposition 3 *The mean-variance-skewness frontier of $\mathbf{r}^{\mathbf{x}}$ spans the mean-variance-skewness frontier of $(\mathbf{r}^{\mathbf{x}}, \mathbf{r}^{\mathbf{y}})$ with short-sales constraints on benchmark and test assets, if*

$$\boldsymbol{\mu}_{\mathbf{y}} - \boldsymbol{\Sigma}_{\mathbf{y}\mathbf{x}^j} \boldsymbol{\Sigma}_{\mathbf{x}^j\mathbf{x}^j}^{-1} \boldsymbol{\mu}_{\mathbf{x}^j} \leq \mathbf{0}_n, \quad (8)$$

$$\boldsymbol{\Sigma}_{\mathbf{y}\mathbf{x}^j} \boldsymbol{\Sigma}_{\mathbf{x}^j\mathbf{x}^j}^{-1} \mathbf{1}_{L^j} - \mathbf{1}_n \leq \mathbf{0}_n, \quad (9)$$

$$\mathbf{S}_{\mathbf{y}\mathbf{x}^j\mathbf{x}^j} - \boldsymbol{\Sigma}_{\mathbf{y}\mathbf{x}^j} \boldsymbol{\Sigma}_{\mathbf{x}^j\mathbf{x}^j}^{-1} \mathbf{S}_{\mathbf{x}^j\mathbf{x}^j\mathbf{x}^j} \leq \mathbf{0}_{n \times (L^j)^2}, \quad (10)$$

for $j = 1, \dots, M$ and where the inequalities apply element-wise.

Proof. See Appendix B. ■

In the empirical section, we will test these restrictions with Wald tests for inequality constraints [Gourieroux et al., 1982].

2.2.2 Spanning and intersection with a risk-free asset

So far, we have presented the general case without a risk-free asset which can be relevant even when a risk-free asset is available if the investor's horizon exceeds the maturity of the risk-free asset [Bajoux-Besnainou et al., 2001] or in an analysis with real returns, especially with a long investment horizon [Chapter 4 of Campbell and Viceira, 2002]. On short horizons investors can usually borrow and invest at a risk-free rate, i.e., the Lagrange multiplier η is zero. Let the superscript e denote

returns in excess of the risk-free rate. The condition for mean-variance-skewness intersection and spanning with a risk-free asset follow almost directly from Proposition 1 and 2 and are given in the next two corollaries.

Corollary 1 *Suppose a risk-free asset is available. The mean-variance-skewness frontier of $\mathbf{r}^x$ intersects the mean-variance-skewness frontier of $(\mathbf{r}^x, \mathbf{r}^y)$, if*

$$\boldsymbol{\mu}_y^e - \boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\boldsymbol{\mu}_x^e + \gamma_2 \{ \mathbf{S}_{yxx} - \boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\mathbf{S}_{xxx} \} (\mathbf{w}_x^* \otimes \mathbf{w}_x^*) = \mathbf{0}_n. \quad (11)$$

Corollary 2 *Suppose a risk-free asset is available. The mean-variance-skewness frontier of $\mathbf{r}^x$ spans the mean-variance-skewness frontier of $(\mathbf{r}^x, \mathbf{r}^y)$, if*

$$\boldsymbol{\mu}_y^e - \boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\boldsymbol{\mu}_x^e = \mathbf{0}_n, \quad (12)$$

$$\mathbf{S}_{yxx} - \boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\mathbf{S}_{xxx} = \mathbf{0}_{n \times k^2}. \quad (13)$$

If there are short-sales constraints, it is straightforward to adapt the general case of short-sales constraints without a risk-free asset to the case with a risk-free asset by testing (8) and (10) for excess returns over the risk-free rate.

3 Empirical results

3.1 Data

The analysis takes the perspective of investors who are able to invest in bonds and stocks and possibly in a risk-free asset¹². The proxy for the performance of stocks is the return on the S&P 500 total return index from Morningstar (henceforth, stocks) and the proxy for bonds is the return

¹²See, e.g., Ait-Sahalia and Brandt [2001] and Kan and Zhou [2012] for studies which use the same set of benchmark assets.

on the 10-year US treasury bond index from CRSP (henceforth, bonds). The proxy for the risk-free rate is the return on the 30-day t-bill index from CRSP.

The potential additional assets are net of fee returns on hedge funds and mutual funds from the Morningstar database. The mutual funds are the live and dead open-end funds which are classified as US equity by the US Category Group and have the US Dollar as a base currency and at least 36 months of return history available. For each fund identifier, we only keep the share class with the longest return history.

The hedge funds are live and dead funds which have the USD as a base currency. We start with all funds available in the database and then apply the following standard filter. First, we delete the first 12 month of return data of each fund to mitigate the instant history or back-fill bias [Fung and Hsieh, 2001, Bali et al., 2013]. Second, each fund is required to have at least 36 months of return data available to have enough data at the individual fund level for our analysis [Bali et al., 2013, Patton and Ramadorai, 2013]. Finally and to avoid having very similar investments multiple times, we keep for each strategy identifier the fund with the longest return history. Applying all these filters reduces our sample from initially 9,717 funds to 4,753 funds (3,021 liquidated, 75 merged and 1,657 live). The data ranges from January 1993 to December 2014 but effectively the data starts in January 1994 after the first twelve months of returns for each fund have been deleted. We use unsmoothed hedge fund returns, i.e., returns adjusted such that their first-order autocorrelation is zero [Getmansky et al., 2004]. Adjusting for smoothing does not change the mean returns but increases the average volatility in our dataset from 3.61% to 4.28% per month. Hence, using unsmoothed returns makes our results more conservative, and we verify in the robustness section that our results are robust to using raw returns.

Morningstar classifies hedge funds in six broad categories¹³: directional debt, directional equity,

¹³See Morningstar [2011] for details.

event, global derivatives, multistrategy and relative value. Throughout the analysis, we essentially keep the Morningstar classification except that we associate fund of funds, which are included in the multistrategy category by Morningstar, to a distinct fund of funds category. Funds without any Morningstar classification are associated to an additional category labeled other. To get an idea of the performance of the different categories, we analyze net asset value weighted portfolios of hedge funds in each category in addition to individual funds.

The descriptive statistics of the monthly excess returns of the benchmark assets and the portfolios of hedge funds are reported in Panel A of Table 1. Panels B to D of the table contain the percentiles of the cross-sectional distribution of the summary statistics of the excess returns and the characteristics of the individual funds.

[Insert Table 1 here.]

Panel A shows that the average monthly excess returns on stocks and bonds are 0.63% and 0.28% and their standard deviations are 4.32% and 2.03%, respectively, over the sample period. Portfolios invested in directional debt, event and multistrategy have a similar average excess return and a lower standard deviation than stocks and are therefore likely to provide a better mean-variance trade-off than stocks. Portfolios invested in multistrategy and relative value have a higher average excess return and a lower standard deviation than bonds. Panel A also shows that all hedge fund strategies have significantly skewed returns. The critical value at a 5% significance level for the null hypothesis that skewness is zero for a sample size of 252 observations is approximately 0.30 under the normal distribution, and the skewness of every strategy exceeds that threshold in absolute value. The strategies also exhibit significant kurtosis as the critical value for the null of kurtosis equal to 3 is approximately 3.60 under the normal distribution. Although kurtosis is statistically significant, we will show in Section 3.4.2 that the effect of kurtosis on utility is economically small. In terms of number of funds in each category, the directional equity category contains the largest number of

funds followed by the fund of funds category. All categories contain more than 200 funds.

Portfolios of funds understate the standard deviation of hedge fund returns as the average standard deviation is 4.27% in Panel B, which is more than twice as large as the standard deviation of the portfolios of hedge funds. Individual hedge funds have slightly lower average excess returns than mutual funds but also lower standard deviations. Notice as well that the dispersion in the cross-sectional distribution of skewness of individual hedge fund returns is larger than the dispersion in skewness of mutual funds. Finally, our hedge funds have typical hedge fund features such as rather few assets under management, substantial management and performance fees, withdrawal restrictions and high minimum investment requirements.

3.2 Portfolios of hedge funds

3.2.1 Intersection

The summary statistics suggest that some portfolios of hedge funds provide substantial diversification benefits to the bond and stock investors. Formal tests of the hypothesis of mean-variance-skewness intersection are reported in Table 2. We consider two cases: $\gamma_1 = 5$ and $\gamma_2 = 15$ (i.e., an investor with relatively low aversion to variance and high preference for skewness, investor A) in column 1-4 and $\gamma_1 = \gamma_2 = 15$ (i.e., an investor with higher aversion to variance and high preference for skewness, investor B) in column 5-8.

[Insert Table 2 here.]

Suppose first that investors can borrow and invest at the risk-free rate. If the investment universe contains only the benchmark assets, investor A allocates 0.73 to stocks and 1.55 to bonds, and investor B allocates 0.26 to stocks and 0.54 to bonds. The intersection tests with a risk-free asset depend indirectly on γ_1 through the optimal portfolio weights which determine the magnitude of

co-skewness. For example, the directional debt Wald statistic equals 2.76 for investor A, who holds more risky assets, and 3.79 for investor B. Notice that directional debt has negative co-skewness with the benchmark assets, and the effect of co-skewness is small for investor B as 3.79 is close to the Wald statistic of the corresponding mean-variance spanning test in Table 3 (3.97). Overall, the table shows that event, global derivatives, multistrategy, and relative value provide significant¹⁴ diversification benefits for both investors with and without short-sales constraints.

Without a risk-free asset exactly the entire portfolio has to be invested and investor A allocates 0.48 to stocks and 0.52 to bonds whereas investor B allocates 0.30 to stocks and 0.70 to bonds. The investors now have very different benefits from the hedge fund portfolios. For investor B, all funds except the category other provide significant diversification benefits with and without short-sales constraints because the hedge fund portfolios enable him to reduce the variance of his portfolio. For investor A instead, who used to borrow at the risk-free rate and invest more than his total wealth in the risky assets, only directional debt, event and multistrategy provide significant diversification benefits.

The case of intersection takes the perspective of specific investors, and intersection for several investors does not imply spanning in the mean-variance-skewness case. This is because there is no fund separation for mean-variance-skewness utility except under further assumptions on the return distribution [Mencia and Sentana, 2009] or approximations of portfolio weights [Chabi-Yo et al., 2014]. Next, we consider the more general case of spanning, which involves statements for all levels of aversion to variance and preference for skewness.

¹⁴Throughout the paper "significant" means significant at the 5% level.

3.3 Spanning

We consider both mean-variance and mean-variance-skewness spanning and report the results in Table 3. The table contains the case with and without a risk-free asset and the case with and without short-sales.

[Insert Table 3 here.]

Mean-variance spanning with a risk-free asset is rejected for portfolios of directional debt, event, global derivatives, multistrategy, and relative value. The evidence against mean-variance spanning becomes stronger when short-sales constraints are taken into account and/or a risk-free asset is not available. The explanation for the former result is that mean-variance investors optimally buy hedge funds and for the latter result that hedge funds enable investors to construct portfolios with lower variance¹⁵. For the mean-variance-skewness case, there is strong evidence against spanning without short-sales constraints, but the evidence is much weaker with short-sales constraints. Spanning is then only rejected for event, global derivatives, multistrategy, and relative value. Hence, the benefits of the portfolio of, e.g., directional debt, in terms of skewness can only be achieved with short-positions, and it is especially important to take short-sales into account when testing for mean-variance-skewness spanning.

On the bottom line, portfolios of event, global derivatives, multistrategy and relative value funds seem to offer robust mean-variance and mean-variance-skewness benefits for investors. The analysis, however, relies on portfolios of hedge funds and as shown in the descriptive statistics, portfolios understate the standard deviation of individual returns and may therefore overstate the benefits of hedge fund investments. The next section uses returns on individual funds to assess the

¹⁵The spanning test without a risk-free asset and the test with a risk-free asset differ mainly because the former test imposes an additional restriction on the sum of betas. As discussed in Kan and Zhou [2012], the condition on the sum of betas measures the change in the global minimum variance portfolios of benchmark assets only and benchmark and test assets. Hence, the higher number of rejection of spanning without risk-free asset is mainly driven by the effect of hedge funds on the global minimum variance portfolio.

benefits of investing in hedge funds.

3.4 Individual funds

3.4.1 Spanning

Below we focus on spanning with a risk-free asset which has the advantage that rejections of spanning are driven by differences in tangency portfolios and not by the economically less meaningful differences in global minimum variance portfolios [Kan and Zhou, 2012]. To analyze the different impact of the restrictions on the intercepts and the residual co-skewnesses, we sort funds independently in quintiles according to the intercept of the mean-variance regression and quintiles according to the residual co-skewness Wald statistic¹⁶. We then report the average rejection rates at the 5% significance level of the null hypothesis of spanning within these quintiles and the intersection of these quintiles. The results are reported in Table 4.

[Insert Table 4 here.]

The table confirms the idea that it is important to look at funds individually because portfolios of hedge funds overstate the diversification benefits of hedge fund investments. While mean-variance spanning was rejected for more than half of the portfolios of hedge funds, mean-variance spanning is rejected only for 22% of the individual funds without short-sales constraints. Similarly, mean-variance-skewness spanning is rejected for 38% of the individual funds whereas it was rejected for half of the portfolios of hedge funds. With short-sales constraints the rejection rate of mean-variance spanning increases to 27% and the rejection rate of mean-variance-skewness spanning decreases to 32%. Hence, mean-variance benefits are usually attainable with long positions in hedge funds while benefits in terms of skewness often involve short positions. In addition, in the second alpha quintile

¹⁶Note that there are three residual co-skewnesses for each fund, one of the fund with stocks, one with bonds and one with the product of bond and stock returns. Therefore, it is necessary to aggregate the co-skewnesses to one variable for the sorts, and the Wald statistics are a natural choice.

in which alpha is close to zero and mean-variance spanning is rejected less than 1% of the time, mean-variance-skewness spanning is still rejected 21% of the time. Likewise, in the first and second residual co-skewness Wald statistic quintile in which the residual co-skewness Wald statistic is close to zero, mean-variance-skewness spanning is rejected 12% and 16% of the time respectively. This shows that funds which provide no benefits in terms of either mean-variance trade-off or skewness may still provide significant benefits in terms of mean-variance-skewness.

To get qualitative predictions about which hedge fund characteristics are related to a high spanning test statistic, we run cross-sectional regressions of first the t-statistic of the intercept of the mean-variance regression on fund characteristics and then the logarithm of one plus the Wald statistic of the mean-variance-skewness spanning test¹⁷ with short-sales constraints on fund characteristics. The results are reported in Table 5.

[Insert Table 5 here.]

Funds which offer mean-variance and mean-variance-skewness benefits to short-sales constrained investors tend to be smaller at inception and have higher performance fees, higher minimum investment requirements, longer lockup and advance notice periods, and their returns have lower standard deviations and higher skewness. In terms of investment strategy, directional debt, event, and multistrategy funds provide significantly more mean-variance benefits compared to directional equity funds. For mean-variance-skewness spanning with short-sales constraints, directional debt and multistrategy funds have higher Wald statistics and fund of funds have significantly lower Wald statistics. The latter result is in line with Brown et al. [2012] who show that fund of funds tend to have lower returns which may be related to the fact that they add another layer of costs to investors by construction.

¹⁷We take the logarithm because the distribution of these Wald statistics has a very long right tail and add one because some Wald statistics are equal to zero.

Our analysis so far has relied on rejection rates to assess the diversification of individual hedge funds and the importance of higher moments. But it is hard to assess whether or not the rejection rates of spanning for hedge funds are high without a reasonable comparison. Therefore, we apply the analysis in Table 4 to our U.S. equity mutual funds sample and report the results in Table 6. The results show that the fraction of mutual funds which deliver significant diversification benefits is smaller than the corresponding fraction of hedge funds. Mean-variance spanning is only rejected for 15% of the funds and mean-variance-skewness spanning for 21% of the funds. Short-sales constraints reduce the rejection rates to 9% for both mean-variance and mean-variance-skewness spanning. Hence, for mutual funds the mean-variance and skewness benefits often involve short positions in the funds. In addition, in the fourth alpha quintile in which mean-variance spanning is rejected for less than 1% of the funds, mean-variance-skewness is only rejected for 9% of the funds whereas the corresponding value for hedge funds is 21%. Overall, (co-)skewness seems to matter for mutual funds as well but less than for hedge funds. In addition, the fraction of funds which significantly improve the portfolios for mean-variance and mean-variance-skewness investors is smaller for mutual funds.

[Insert Table 6 here.]

3.4.2 Performance analysis

Leaving the mean-variance framework with a risk-free asset means that whether or not an asset improves the optimal portfolio for all investors can no longer be evaluated with a single measure, alpha. To extend alpha to take into account higher moments it is necessary to further specify the risk preferences of the investor. This corresponds to the case of intersection and we can define the

prudent alpha, i.e., the alpha adjusted for skewness, as¹⁸

$$\alpha_i^{pr} = \alpha_i + \gamma_2 E \left[\epsilon_i (r_p - \mu_p)^2 \right], \quad (14)$$

where α_i is the intercept of the linear regression of asset i 's return on the benchmark asset returns, i.e., the simple alpha, ϵ_i is the residual of that regression, and r_p is the return on the optimal benchmark portfolio of the investor. Notice that if there is intersection for the individual with preferences (γ_1, γ_2) who optimally holds portfolio r_p , then $\alpha_i^{pr} = 0$. If there is no intersection, α_i^{pr} measures the additional return the asset delivers in excess of the return required by the investor.

The comparison of simple and prudent alphas gives the magnitude of the impact of skewness in terms of utility cost. In addition, the prudent alpha can be compared to the generalized alpha, i.e., the alpha adjusted for all higher moments, and the kurtosis adjusted alpha to assess how important the skewness adjustment is. The generalized alpha for a CRRA utility function is [Leland, 1999]

$$\alpha_i^{CRRA} = E[r_i^e] - \beta_i^{CRRA} E[r_p^e], \quad (15)$$

where $\beta_i^{CRRA} = \text{Cov}(r_i, (1 + r_p)^{-\gamma}) / \text{Cov}(r_p, (1 + r_p)^{-\gamma})$ and γ is the coefficient of relative risk aversion of the investor. Below we will also compare the impact of skewness to the impact of the next moment, kurtosis. The alpha adjusted for skewness and kurtosis will be referred to as the temperant alpha¹⁹ and it can be obtained in the same way as the prudent alpha. It equals

$$\alpha_i^{te} = \alpha_i^{pr} - \gamma_3 E \left[\epsilon_i (r_p - \mu_p)^3 \right], \quad (16)$$

¹⁸To obtain the expression for prudent alpha, assume that (11) is not equal to zero but equal to α_i^{pr} for test asset i . Then notice that $\alpha_i = \boldsymbol{\mu}_{y_i}^e - \boldsymbol{\Sigma}_{y_i x} \boldsymbol{\Sigma}_{xx}^{-1} \boldsymbol{\mu}_x^e$ and $E[\epsilon_i (r_p - \mu_p)^2] = \{\boldsymbol{S}_{y_i xx} - \boldsymbol{\Sigma}_{y_i x} \boldsymbol{\Sigma}_{xx}^{-1} \boldsymbol{S}_{xx x}\} (\boldsymbol{w}_x^* \otimes \boldsymbol{w}_x^*)$.

¹⁹Temperance is a term coined by Kimball [1992] to refer to a negative fourth derivative of the utility function which measures the aversion to the fourth moment.

where γ_3 is a positive scalar which measures the aversion to kurtosis (relative to the preference for the mean)²⁰.

Next we calculate these adjusted alphas for the individual funds to assess the magnitude of the effect of higher moments on required returns. The adjusted alphas are calculated for a CRRA utility function with relative risk aversion of 10²¹ which implies $\gamma_1 = \gamma/(1 + \mu_p)$ and $\gamma_2 = \frac{1}{2}\gamma(\gamma + 1)/(1 + \mu_p)^2 = 55/(1 + \mu_p)^2$. For each fund the adjustment is calculated for a portfolio of stocks and bonds, which is optimally chosen to maximize the average CRRA utility over the sample period. The results are reported in Table 7. To maximize the cross-sectional dispersion, we sort the funds in alpha quintiles and residual co-skewness quintiles and further split each lowest and highest quintile into two deciles. The table reports the annualized average of α_i , $\alpha_i^{pr} - \alpha_i$, $\alpha_i^{te} - \alpha^{pr}$ and $\alpha_i^{CRRA} - \alpha^{pr}$.

[Insert Table 7 here.]

The average annualized simple alpha of hedge funds is 3.3% and alphas range from -9.7% in the lowest alpha decile to around 17.3% in the highest alpha decile. While the average prudent and simple alphas are similar, the differences between the prudent and simple alphas are large in the cross-section. In the lowest residual co-skewness decile, simple alphas overestimate the added value of hedge funds by about 1.2% per year on average and in the highest residual co-skewness decile simple alphas under-estimate the benefits of hedge funds by about 1.2% per year on average. Hence, the cost associated to the use of mean-variance performance metrics adds up to 2.4% in the cross-section. The results in Table 7 also suggest that the cost of ignoring the moments beyond the third-order is small. The difference between prudent and generalized alphas, which take into account all higher moments, are very small and, on average, never exceed 0.15% in absolute value

²⁰Following the approach of Appendix A, γ_3 equals $-\frac{1}{6} \frac{u''''(\cdot)}{u'(\cdot)}$ in an expected utility framework.

²¹A relative risk aversion of 10 is considered to be in the range of reasonable values of risk aversion and yields realistic portfolio allocations. We show in Section 3.6 that our results are robust to using other values of relative risk aversion.

per year. In addition, the difference between prudent and temperant alphas does not exceed 0.19% on average in absolute value per year suggesting that the next moment, kurtosis, has less economic significance. Overall, these results support the finding that the mean-variance-skewness framework captures the most meaningful difference between the mean-variance framework and a more specific framework with all higher moments.

The next table relates simple alpha and the residual co-skewness of Table 7 to fund characteristics. Simple alphas are positively related to management and performance fees, which is in line with the intuition that good funds are able to charge higher fees and that performance fees better align the incentive of managers and investors. In addition, simple alphas are positively related to minimum investment requirements, which is in line with Aragon [2007] who shows that performance is positively related to share restrictions. Residual co-skewness appears to be different from alpha as it is not related to compensation and negatively related to the length of the sum of the lockup period and the advance notice days. This negative relation may be in line with the limits to arbitrage model of Hombert and Thesmar [2014] in the sense that hedge funds which are protected against outflows optimally take more downside risk in the form of negative residual co-skewness. Moreover, residual co-skewness is larger for funds with more volatile returns. In terms of investment strategy and compared to directional equity funds, directional debt, event and multistrategy funds have higher alphas, and directional debt and other funds have lower residual co-skewness.

[Insert Table 8 here.]

For comparison, we also report α , $\alpha^{pr} - \alpha$, $\alpha^{te} - \alpha^{pr}$ and $\alpha^{CRRR} - \alpha^{pr}$ for the mutual fund sample in Table 6. The average simple alpha of mutual funds is almost zero consistent with the notion that these funds do not add value to investors on average. The skewness adjustment adds up to about 0.85% per year in the cross-section of mutual funds. Hence, using prudent alphas is less important for mutual funds than it is for hedge funds. Adjusting further for kurtosis and all

higher moments changes alphas in absolute value by up to 0.09% and 0.13%, respectively.

[Insert Table 9 here.]

3.5 Robustness

Having shown that rejection rates and performance adjustments are considerably higher for hedge funds than for mutual funds, we investigate in this subsection the robustness of this result to using subsets of hedge funds and alternative modeling assumptions.

3.5.1 Spanning for different fund types

Table 10 reports the rejection rates of the null hypothesis of spanning for various sub-samples. The insights can be summarized as follows. First, rejection rates are very similar for small and large funds, as measured by the funds first net asset value. Second, older funds, i.e., funds with longer return histories, tend to perform better by mean-variance metrics whereas younger funds are more interesting from a skewness perspective. Third, over time the fraction of funds which provide mean-variance spanning benefits has halved while the fraction of funds which provide mean-variance-skewness benefits has decreased less. This suggests that skewness is especially important in the recent subsample. Fourth, rejection rates within Morningstar categories are mostly consistent with the results on portfolios in Table 3, i.e., rejection rates for mean-variance-skewness spanning with short-sales constraints are the highest for directional debt, event, global derivatives, relative value and multistrategy category. Fifth, using simple returns (instead of smoothing adjusted returns) or deleting the first 24 returns for each fund (instead of deleting only the first 12 returns) to adjust for back-fill bias yields very similar results. Finally, including delisting returns for liquidated funds²² of -25% and -50% as suggested by Aiken et al. [2013] reduces the rejection rates considerably. Overall,

²²Notice that funds which stop reporting returns to the database are classified as liquidated by Morningstar.

the rejection rates in the main analysis are very similar to the rejection rates in the subsamples.

[Insert Table 10 here.]

3.5.2 Performance analysis for different fund types

Table 11 contains the corresponding robustness checks for the performance analysis. The main insights are the following. First, small funds tend to have higher simple alphas but also larger absolute skewness adjustments (i.e., $|\alpha^{pr} - \alpha|$ is larger). Second, older funds have higher simple alphas and smaller absolute skewness adjustments. Third, simple alphas in the earlier subperiod are considerably higher than in the recent subperiod. At the same time, asymmetries are more important for the recent subperiod. Fourth, in terms of subcategories and in line with earlier results, directional debt has the largest negative skewness adjustments. In addition, global derivatives have the largest absolute skewness adjustments which add up to more than 3% in the cross-section. Fifth, results are very similar for simple returns and when the first 24 returns are deleted. Average alphas are thereby slightly larger in the former case and slightly smaller in the latter case. Sixth, the average simple alpha is close to 0 for a delisting return of -25% and negative at -2.0% for a delisting return of -50% . Seventh, in terms of alternative utility function, skewness adjustments are very similar for CRRA with relative risk aversion of 4. This is consistent with the idea that a lower relative risk aversion decreases the prudence of the investor but increases his optimal portfolio allocation. These two opposing effects approximately cancel out each other in terms of skewness adjustment. In addition, a utility function with a constant habit formation level equal to the average habit level in Campbell and Cochrane [1999] yields similar skewness adjustments as well, i.e., the larger prudence and the smaller optimal portfolio allocation to risky assets approximately offset each other. The moments beyond the fourth-order become more important for the habit formation utility and the total adjustments to alphas, i.e., $\alpha^{CRRA} - \alpha$, add up to more 3% in the cross-section

of funds. Finally, we examine the robustness of the approximation to alternative sorting. Sorting funds into deciles based on residual co-kurtosis yields a kurtosis adjustment, i.e., $\alpha^{te} - \alpha^{pr}$, which adds up to 0.80% in the cross-section. Hence, the economic magnitude of the kurtosis adjustment is considerably smaller than the skewness adjustment. In addition, sorting on $\alpha^{CRRRA} - \alpha$ shows that the skewness adjustment explains almost all the difference between average simple alphas and alphas adjusted for all higher moments.

[Insert Table 11 here.]

3.6 Simulated option strategies

To obtain further insights about how funds can generate option like returns and co-skewness and to compare our method to using factor models with option returns, we assess the performance of option strategies in a simulated economy. As in Leland [1999], the market in this economy follows a geometric Brownian motion with a constant drift and volatility, and the options are priced with the Black-Scholes formulas.

We consider the returns on a buy-write strategy, i.e., a strategy which buys the market and sells a call option, and an insurance strategy which buys the market and a put option. Both strategies are calculated for options ranging from at the money to out of the money, and the options are held to maturity²³. Notice that hedge funds could indeed trade these options or follow dynamic strategies to obtain option-like returns. In addition, the buy-write strategy considered here has a similar payoff as the strategy Jurek and Stafford [2015] propose to replicate hedge fund returns. Their strategy involves selling an out of the money put option and investing a fraction of the strike price at the risk-free rate.

Table 12 shows that the buy-write strategy has a positive simple alpha, and the insurance

²³The results for options strategies sold before maturity are quantitatively and qualitatively very similar and available at request.

strategy has a negative simple alpha. The CRRA alpha of these strategies is zero because as shown in He and Leland [1993] the representative investor in a log normal economy with identically and independently distributed returns must have power utility. Option strategies are then redundant for the representative investor. Table 12 shows as well that the prudent alphas are close to zero and make again up for the largest part of the difference between simple alphas and the alphas adjusted for all higher moments. The table also contains the alphas of strategies with more extreme non-normal returns such as a buy-write strategy which buys the market and sells five call options and the insurance strategy which buys the market and buys five put options. Prudent alphas are still close to zero for these strategies.

[Insert Table 12 here.]

In Table 13 we apply popular factor models to these return series such as the quadratic market model, i.e., a regression on the excess market return and the squared excess market return, and factor models which include the returns on at the money call and put options as factors. None of these models is able to correctly assess that the added value of the strategies is zero. We extend this result in the internet appendix. In particular, Fung and Hsieh [2001] show that lookback straddles explain the returns on trend-following funds well and Agarwal and Naik [2004] use returns on call and put options to explain hedge fund returns. Hence, we have also calculated the returns on straddles and call options in Table 1 of the internet appendix. We find that our framework assigns an alpha close to zero to most of these strategies. In addition, single factor models based on straddles or call options evaluate performance reliably only if the option factors have the same strikes as the assessed option strategies.

[Insert Table 13 here.]

4 Conclusion

Motivated by extensive evidence that hedge fund returns exhibit option-like characteristics, we study whether or not hedge funds improve the investment opportunity set of prudent investors, i.e., investors who are averse to downside risk and have a preference for skewness. To achieve this, we first extend the mean-variance spanning and intersection approach of Huberman and Kandel [1987] to prudent investors. The mean-variance spanning and intersection approach provides tests for the hypothesis that new assets improve the investment opportunity set of mean-variance investors. We derive the conditions under which assets improve the investment opportunity set of prudent investors. We thereby also consider the case of short-sales constraints and discuss implications for performance measurement.

Using a large cross-section of individual hedge funds, we study the importance of option-like characteristics in returns. Our results suggest that prudent investors may benefit from different funds than mean-variance investors. In particular, some hedge funds improve the investment opportunity set of prudent investors although they do not improve the investment opportunity set of mean-variance investors. In addition, we show that prudent investors are willing to forgo almost 2.5% return per year on average to hold a fund which is more attractive from a skewness perspective. Our results on hedge funds contrast with our results on mutual funds for which our conclusions regarding mean-variance and prudent investors differ less and for which a much smaller subset of funds improves the investment opportunity set. We also verify in a simulated economy that our method performs well in determining the alpha of option trading strategies and outperforms widely used option-based factor models.

In summary, our findings highlight the importance of explicitly taking investors' preference for skewness into account when evaluating hedge fund performance from a portfolio perspective. Adding to recent evidence that aggregate hedge fund performance is statistically indistinguishable from put

writing strategies [Jurek and Stafford, 2015], our method suggests that in the cross-section and even after controlling for investors' aversion to downside risk, a subset of hedge funds improves the investment opportunity set of investors.

This paper has focused on the application of our methodology to hedge fund returns. Our method can also be applied to other asset classes. In particular, the returns of currency trading strategies and emerging markets have been in the limelight for their skewness [see, e.g., Brunnermeier et al., 2008, Ghysels et al., 2016, respectively].

A Mean-variance-skewness utility as an approximation to expected utility

The aim of this appendix is to explain how γ_1 and γ_2 are related to the coefficients of risk aversion, i.e., $A(w) \equiv -u''(w)/u'(w)$, and prudence, i.e., $P(w) \equiv -u'''(w)/u''(w)$, of a von Neumann-Morgenstern utility function u . To start take a Taylor approximation of marginal utility derived from end of period wealth around expected wealth²⁴

$$u'(w_0(1+r_p)) = \sum_{i=0}^{\infty} \frac{1}{i!} w_0^i (r_p - \mu_p)^i u^{i+1}(E(\tilde{w})),$$

where $u^i(x)$ is the i th derivative of u with respect to x (i.e., $u^1(\cdot) = u'(\cdot)$, $u^2(\cdot) = u''(\cdot)$, etc.), w_0 is the level of initial wealth, r_p is the portfolio return and μ_p is the expected portfolio return. Truncating the approximation at the second-order and inserting the approximation in the first-order condition of the agent for asset i yields

$$E[r_i u'(w_0(1+r_p))] = E[r_i] u'(\cdot) + u''(\cdot) E[r_i(r_p - \mu_p)] + \frac{1}{2} u'''(\cdot) E[r_i(r_p - \mu_p)^2], \quad (17)$$

where $(\cdot) \equiv w_0(1 + \mu_p)$ and we ignore in (17) that the equalities are indeed approximations. If the first-order condition of the investor holds, the right-hand side of (17) can be rewritten to

$$E[r_i] + \frac{u''(\cdot)}{u'(\cdot)} E[r_i(r_p - \mu_p)] + \frac{1}{2} \frac{u'''(\cdot)}{u'(\cdot)} E[r_i(r_p - \mu_p)^2] = 0, \quad (18)$$

which is very similar to the first-order condition of the investor with mean-variance-skewness utility and, therefore, $\gamma_1 = -\frac{u''(\cdot)}{u'(\cdot)}$ and $\gamma_2 = \frac{1}{2} \frac{u'''(\cdot)}{u'(\cdot)}$. Hence, the aversion to variance relative to the preference for the mean is the absolute risk aversion, i.e., $\gamma_1 = A(\cdot)$, and the preference for skewness

²⁴As a reference and for the technical details see Jurczenko and Maillet [2006] and the references cited therein.

relative to the preference for the mean is one half of the product of risk aversion and prudence, i.e., $\gamma_2 = \frac{1}{2}A(\cdot)P(\cdot)$. Notice that $E[r_i(r_p - \mu_p)]$ is the covariance of asset i with the optimal portfolio of the investor. $E[r_i(r_p - \mu_p)^2]$ is not exactly the co-skewness of asset i with the optimal portfolio because $E[r_i(r_p - \mu_p)^2] - E[(r_i - \mu_i)(r_p - \mu_p)^2] = -\mu_i\sigma_p^2$. Hence, the interpretations of γ_1 and γ_2 in terms of expected utility hold only approximately. However, we show empirically that the alphas adjusted for skewness according to the mean-variance-skewness framework with $\gamma_1 = -\frac{u''(\cdot)}{u'(\cdot)}$ and $\gamma_2 = \frac{1}{2}\frac{u'''(\cdot)}{u'(\cdot)}$ are very close to the alphas adjusted for an expected utility framework. This is consistent with the idea that the mean-variance-skewness framework captures most features of the expected utility framework and that the parameter interpretations are reasonable.

Possible values for γ_1 and γ_2 can now be obtained from standard utility functions like constant absolute risk aversion (CARA) utility functions or constant relative risk aversion (CRRA) utility functions. For CARA of the type $u(x) = -Ae^{-Ax}$ where A is the coefficient of absolute risk aversion, $\gamma_1 = A$ and $\gamma_2 = \frac{1}{2}A^2$. For CRRA of the type $u(x) = x^{1-\gamma}/(1-\gamma)$ with $\gamma \neq 1$, $A(w) = \gamma/w$ and $P(w) = (\gamma+1)/w$ which in turn implies $\gamma_1 = \gamma/w$ and $\gamma_2 = \frac{1}{2}\gamma(\gamma+1)/w^2$. A reasonable value for γ is for example $\gamma = 4$ [see Gollier, 2001], which yields, for $w = 1$, $\gamma_1 = 4$ and $\gamma_2 = 10$. While prudence is a function of risk aversion for CRRA utility functions, hyperbolic absolute risk aversion (HARA) functions, for instance, introduce more flexibility in modeling γ_1 and γ_2 separately. HARA utility is given by $u(z) = \xi(\eta + z/\gamma)^{1-\gamma}$ defined on a domain of z such that $\eta + z/\gamma > 0$ [Gollier, 2001, p. 26] which implies $A(w) = \gamma/(\gamma\eta + w)$ and $P(w) = (\gamma+1)/(\gamma\eta + w)$. A widely used special case of a HARA utility function is the additive habit utility function. Setting $\xi = \left(\gamma^{\frac{1}{1-\gamma}}\right)\left(\frac{1}{1-\gamma}\right)$ and $\eta = -k/\gamma$ in the HARA utility function, we obtain the additive habit utility functions of the form $u(x) = (x - k)^{1-\gamma}/(1-\gamma)$, with k the (constant) habit level and $\gamma \neq 1$. These values imply $A(w) = \frac{\gamma}{w-k}$ and $P(w) = \frac{\gamma+1}{w-k}$. In this case, $\gamma_2 = \frac{1}{2}\gamma_1\left(\gamma_1 + \frac{1}{w-k}\right)$ which allows for a much larger range of combinations of γ_1 and γ_2 . Indeed, low relative risk aversion and very high relative

prudence can be achieved with γ close to zero and k very close to w .²⁵

B Proofs

Proof of Proposition 1. The optimal portfolio $\mathbf{w}^*$ satisfies the first-order conditions

$$\begin{pmatrix} \boldsymbol{\mu}_x \\ \boldsymbol{\mu}_y \end{pmatrix} - \gamma_1 \begin{pmatrix} \boldsymbol{\Sigma}_{xx} & \boldsymbol{\Sigma}_{xy} \\ \boldsymbol{\Sigma}_{yx} & \boldsymbol{\Sigma}_{yy} \end{pmatrix} \begin{pmatrix} \mathbf{w}_x^* \\ \mathbf{w}_y^* \end{pmatrix} + \gamma_2 \begin{pmatrix} S_{xxx} & S_{xxy} & S_{xyx} & S_{xyy} \\ S_{yxx} & S_{yxy} & S_{yyx} & S_{yyy} \end{pmatrix} \begin{pmatrix} \mathbf{w}_x^* \otimes \mathbf{w}_x^* \\ \mathbf{w}_x^* \otimes \mathbf{w}_y^* \\ \mathbf{w}_y^* \otimes \mathbf{w}_x^* \\ \mathbf{w}_y^* \otimes \mathbf{w}_y^* \end{pmatrix} - \eta \mathbf{1} = \mathbf{0}, \quad (19)$$

$$\text{and} \quad \mathbf{w}^{*\top} \mathbf{1} - 1 = 0,$$

where the subscripts $\mathbf{x}$ and $\mathbf{y}$ refer to the k benchmark assets and n test assets, respectively, $\mathbf{w}_x^*$ and $\mathbf{w}_y^*$ are the subvectors of $\mathbf{w}^*$, $\boldsymbol{\mu}_x$ and $\boldsymbol{\mu}_y$ are the subvectors of $\boldsymbol{\mu}$, $\boldsymbol{\Sigma}_{xx}$, $\boldsymbol{\Sigma}_{xy}$, $\boldsymbol{\Sigma}_{yx}$ and $\boldsymbol{\Sigma}_{yy}$ are the submatrices of $\boldsymbol{\Sigma}$, S_{xxx} , S_{xxy} , S_{xyx} , S_{xyy} , S_{yxx} , S_{yxy} , S_{yyx} , S_{yyy} are the submatrices of $\mathbf{S}$, and η is the Lagrange multiplier of the budget constraint. If we have mean-variance-skewness intersection (i.e., $\mathbf{w}_y^* = \mathbf{0}$), (19) becomes

$$\begin{pmatrix} \boldsymbol{\mu}_x \\ \boldsymbol{\mu}_y \end{pmatrix} - \gamma_1 \begin{pmatrix} \boldsymbol{\Sigma}_{xx} \mathbf{w}_x^* \\ \boldsymbol{\Sigma}_{yx} \mathbf{w}_x^* \end{pmatrix} + \gamma_2 \begin{pmatrix} S_{xxx} (\mathbf{w}_x^* \otimes \mathbf{w}_x^*) \\ S_{yxx} (\mathbf{w}_x^* \otimes \mathbf{w}_x^*) \end{pmatrix} - \eta \mathbf{1} = \mathbf{0}. \quad (20)$$

²⁵ As an example, take $\gamma = 1/10000$, $w = 1$ and $k = 9999/10000$ which imply $\gamma_1 = 1$ and $\gamma_2 = 5,000.5$.

The first k rows of (20) can then be written as²⁶

$$\mathbf{w}_x^* = \frac{1}{\gamma_1} \Sigma_{xx}^{-1} \boldsymbol{\mu}_x + \frac{\gamma_2}{\gamma_1} \Sigma_{xx}^{-1} \mathbf{S}_{xxx} (\mathbf{w}_x^* \otimes \mathbf{w}_x^*) - \frac{\eta}{\gamma_1} \Sigma_{xx}^{-1} \mathbf{1}_k,$$

and using this to rewrite the last n rows of (20) gives the result of the proposition

$$\boldsymbol{\mu}_y - \Sigma_{yx} \Sigma_{xx}^{-1} \boldsymbol{\mu}_x + \eta (\Sigma_{yx} \Sigma_{xx}^{-1} \mathbf{1}_k - \mathbf{1}_n) + \gamma_2 \{ \mathbf{S}_{yxx} - \Sigma_{yx} \Sigma_{xx}^{-1} \mathbf{S}_{xxx} \} (\mathbf{w}_x^* \otimes \mathbf{w}_x^*) = \mathbf{0}_n. \quad (21)$$

■

Proof of Proposition 2. Mean-variance-skewness spanning requires that (21) is satisfied for all values of γ_2 and η (and associated $\mathbf{w}_x^*$). Hence, the conditions for mean-variance-skewness spanning are

$$\boldsymbol{\mu}_y - \Sigma_{yx} \Sigma_{xx}^{-1} \boldsymbol{\mu}_x = \mathbf{0}_n,$$

$$\Sigma_{yx} \Sigma_{xx}^{-1} \mathbf{1}_k - \mathbf{1}_n = \mathbf{0}_n,$$

$$\mathbf{S}_{yxx} - \Sigma_{yx} \Sigma_{xx}^{-1} \mathbf{S}_{xxx} = \mathbf{0}_{n \times k^2},$$

where the equalities apply element-wise. ■

Proof of Proposition 3. The portfolio problem with short-sales constraints is

$$\begin{aligned} \max_{\mathbf{w}} \quad & \mathbf{w}^\top \boldsymbol{\mu} - \frac{1}{2} \gamma_1 \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w} + \frac{1}{3} \gamma_2 \mathbf{w}^\top \mathbf{S} (\mathbf{w} \otimes \mathbf{w}), \\ \text{s.t.} \quad & \mathbf{w}^\top \mathbf{1} = 1 \quad \text{and} \quad w_i \geq 0, \forall i. \end{aligned}$$

²⁶Note that the mean-variance-skewness portfolio problem has no closed form solution for portfolio weights. In addition, there is no three fund separation for arbitrary distributions because it is not possible to write the optimal portfolio of any investor as a function of three distinct funds. Three fund separation can be obtained with additional distributional assumptions as for example in Mencia and Sentana [2009].

Let the vector $\boldsymbol{\delta}$ contain the Kuhn-Tucker multipliers for the restriction that the portfolio weights are non-negative. The mean-variance-skewness efficient portfolio $\mathbf{w}^*$ satisfies

$$\boldsymbol{\mu} - \eta \mathbf{1} + \boldsymbol{\delta} = \gamma_1 \boldsymbol{\Sigma} \mathbf{w}^* - \gamma_2 \mathbf{S}(\mathbf{w}^* \otimes \mathbf{w}^*), \quad (22)$$

$$w_i^*, \delta_i \geq 0, \quad \forall i,$$

$$w_i^* \delta_i = 0, \quad \forall i,$$

$$\mathbf{w}^{*\top} \mathbf{1} = 1.$$

If we have mean-variance-skewness intersection (i.e., $\mathbf{w}_y^* = \mathbf{0}$), (22) can be rewritten to

$$\begin{pmatrix} \boldsymbol{\mu}_x \\ \boldsymbol{\mu}_y \end{pmatrix} - \gamma_1 \begin{pmatrix} \boldsymbol{\Sigma}_{xx} \mathbf{w}_x^* \\ \boldsymbol{\Sigma}_{yx} \mathbf{w}_x^* \end{pmatrix} + \gamma_2 \begin{pmatrix} \mathbf{S}_{xxx}(\mathbf{w}_x^* \otimes \mathbf{w}_x^*) \\ \mathbf{S}_{yxx}(\mathbf{w}_x^* \otimes \mathbf{w}_x^*) \end{pmatrix} + \boldsymbol{\delta} - \eta \mathbf{1} = \mathbf{0}. \quad (23)$$

We proceed as DeRoos et al. [2001] and take the mean-variance-skewness efficient portfolio which implies a particular value of η . Let $\mathbf{r}^{x^\eta}$ refer to the L -dimensional subvector of $\mathbf{r}^x$ which contains only the returns of the assets for which short-sales constraints are not binding and let superscripts $^\eta$ refer to this subset. (23) becomes then

$$\boldsymbol{\mu}_{x^\eta} - \gamma_1^\eta \boldsymbol{\Sigma}_{x^\eta x^\eta} \mathbf{w}^\eta + \gamma_2^\eta \mathbf{S}_{x^\eta x^\eta x^\eta}(\mathbf{w}^\eta \otimes \mathbf{w}^\eta) - \eta \mathbf{1}_L = \mathbf{0}_L, \quad (24)$$

and

$$\begin{pmatrix} \boldsymbol{\mu}_x \\ \boldsymbol{\mu}_y \end{pmatrix} - \gamma_1^\eta \begin{pmatrix} \boldsymbol{\Sigma}_{xx^\eta} \mathbf{w}^\eta \\ \boldsymbol{\Sigma}_{yx^\eta} \mathbf{w}^\eta \end{pmatrix} + \gamma_2^\eta \begin{pmatrix} \mathbf{S}_{xx^\eta x^\eta}(\mathbf{w}^\eta \otimes \mathbf{w}^\eta) \\ \mathbf{S}_{yx^\eta x^\eta}(\mathbf{w}^\eta \otimes \mathbf{w}^\eta) \end{pmatrix} + \boldsymbol{\delta} - \eta \mathbf{1} = \mathbf{0}.$$

Using (24) we get the condition on the test assets for intersection

$$\begin{aligned} & \boldsymbol{\mu}_y - \boldsymbol{\Sigma}_{yx^\eta} \boldsymbol{\Sigma}_{x^\eta x^\eta}^{-1} \boldsymbol{\mu}_{x^\eta} + \eta (\boldsymbol{\Sigma}_{yx^\eta} \boldsymbol{\Sigma}_{x^\eta x^\eta}^{-1} \mathbf{1}_k - \mathbf{1}_n) \\ & + \gamma_2^\eta \{ \boldsymbol{S}_{yx^\eta x^\eta} - \boldsymbol{\Sigma}_{yx^\eta} \boldsymbol{\Sigma}_{x^\eta x^\eta}^{-1} \boldsymbol{S}_{x^\eta x^\eta x^\eta} \} (\boldsymbol{w}^\eta \otimes \boldsymbol{w}^\eta) + \boldsymbol{\delta}_n = \mathbf{0}_n, \end{aligned}$$

or

$$\begin{aligned} & \boldsymbol{\mu}_y - \boldsymbol{\Sigma}_{yx^\eta} \boldsymbol{\Sigma}_{x^\eta x^\eta}^{-1} \boldsymbol{\mu}_{x^\eta} + \eta (\boldsymbol{\Sigma}_{yx^\eta} \boldsymbol{\Sigma}_{x^\eta x^\eta}^{-1} \mathbf{1}_{k^\eta} - \mathbf{1}_n) \\ & + \gamma_2^\eta \{ \boldsymbol{S}_{yx^\eta x^\eta} - \boldsymbol{\Sigma}_{yx^\eta} \boldsymbol{\Sigma}_{x^\eta x^\eta}^{-1} \boldsymbol{S}_{x^\eta x^\eta x^\eta} \} (\boldsymbol{w}^\eta \otimes \boldsymbol{w}^\eta) \leq \mathbf{0}_n. \end{aligned} \quad (25)$$

Spanning implies that (25) holds for all relevant values of η and γ_2^η . Again, we follow the exposition of DeRoos et al. [2001] to give the conditions for spanning. Let $\mathcal{H}^j$ and Γ^j be the sets of η and γ_2^η , respectively, for which the subset of assets for which the short-sales constraints in the mean-variance-skewness efficient portfolios are not binding is the same. In addition, let the L^j -dimensional vector of returns of these assets be denoted as $\boldsymbol{r}^{x^j}$, i.e., $\boldsymbol{r}^{x^j} = \boldsymbol{r}^{x^\eta}$ if and only if $\eta \in \mathcal{H}^j$ and $\gamma_2^\eta \in \Gamma^j$. As before, each variable which refers to the set $\boldsymbol{r}^{x^j}$, $j = 1, 2, \dots, M$, is denoted with a superscript j . Hence, we have mean-variance-skewness spanning if and only if the M conditions,

$$\begin{aligned} & \underbrace{\boldsymbol{\mu}_y - \boldsymbol{\Sigma}_{yx^j} \boldsymbol{\Sigma}_{x^j x^j}^{-1} \boldsymbol{\mu}_{x^j} + \eta (\boldsymbol{\Sigma}_{yx^j} \boldsymbol{\Sigma}_{x^j x^j}^{-1} \mathbf{1}_{L^j} - \mathbf{1}_n)}_A \\ & + \underbrace{\gamma_2^\eta \{ \boldsymbol{S}_{yx^j x^j} - \boldsymbol{\Sigma}_{yx^j} \boldsymbol{\Sigma}_{x^j x^j}^{-1} \boldsymbol{S}_{x^j x^j x^j} \}}_B (\boldsymbol{w}^\eta \otimes \boldsymbol{w}^\eta) \leq \mathbf{0}_n, \end{aligned} \quad (26)$$

$\forall \eta \in \mathcal{H}^j$ and $\forall \gamma_2^\eta \in \Gamma^j$, hold. Note that a sufficient condition for part B of (26) to be non-positive is that all elements of $\boldsymbol{S}_{yx^j x^j} - \boldsymbol{\Sigma}_{yx^j} \boldsymbol{\Sigma}_{x^j x^j}^{-1} \boldsymbol{S}_{x^j x^j x^j}$ are non-positive because γ_2^η is non-negative and all elements of $\boldsymbol{w}^\eta$ are positive. In addition, denoting $\eta_{\inf}^j = \inf(\mathcal{H}^j)$ and $\eta_{\sup}^j = \sup(\mathcal{H}^j)$, it is

sufficient for part A of (26) to be non-positive if A is non-positive for $\eta_{\inf}^j$ and $\eta_{\sup}^j$ because it is then non-positive $\forall \eta \in \mathcal{H}^j$. These conditions taken together are

$$\begin{aligned}\mu_y - \Sigma_{yx^j} \Sigma_{x^j x^j}^{-1} \mu_{x^j} + \eta_{\inf}^j (\Sigma_{yx^j} \Sigma_{x^j x^j}^{-1} \mathbf{1}_{L^j} - \mathbf{1}_n) &\leq \mathbf{0}_n, \\ \mu_y - \Sigma_{yx^j} \Sigma_{x^j x^j}^{-1} \mu_{x^j} + \eta_{\sup}^j (\Sigma_{yx^j} \Sigma_{x^j x^j}^{-1} \mathbf{1}_{L^j} - \mathbf{1}_n) &\leq \mathbf{0}_n, \\ S_{yx^j x^j} - \Sigma_{yx^j} \Sigma_{x^j x^j}^{-1} S_{x^j x^j x^j} &\leq \mathbf{0}_{n \times (L^j)^2},\end{aligned}$$

for $j = 1, \dots, M$. A lower bound on η is obtained by not imposing the condition that all wealth has to be invested, i.e., $0 \leq \mathbf{w}^\top \mathbf{1} \leq 1$, which implies that $\eta \in [0, +\infty)$. Sufficient conditions for mean-variance-skewness spanning without short-sales are then

$$\begin{aligned}\mu_y - \Sigma_{yx^j} \Sigma_{x^j x^j}^{-1} \mu_{x^j} &\leq \mathbf{0}_n, \\ \Sigma_{yx^j} \Sigma_{x^j x^j}^{-1} \mathbf{1}_{L^j} - \mathbf{1}_n &\leq \mathbf{0}_n, \\ S_{yx^j x^j} - \Sigma_{yx^j} \Sigma_{x^j x^j}^{-1} S_{x^j x^j x^j} &\leq \mathbf{0}_{n \times (L^j)^2},\end{aligned}$$

for $j = 1, \dots, M$. ■

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Table 1: Descriptive statistics

The table shows the summary statistics. Panel A contains the number of observations (T), average (mean), standard deviation (std), skewness (skew), kurtosis (kurt), p-value of the Jarque-Bera test in the interval $[0.001, 0.5]$, minimum (min) and maximum (max) of the monthly returns in excess of the risk-free rate on the benchmark assets (stocks and bonds) and portfolios invested in hedge funds within a Morningstar category and across the whole hedge fund universe. Each fund in these portfolios is weighted by its net asset value at the end of the previous month and at least 10 funds are required to report returns and net asset values in a given month to calculate a portfolio return. The number of distinct funds in each Morningstar category is reported in the last column of the table. The proxy for the stock and bond markets are the S&P 500 TR index and 10 year US T-Bonds, respectively, and the return on the 30 day T-Bill proxies for the risk-free rate. Panel B contains the percentiles of the summary statistics of individual hedge funds. The fund size is reported for 1,619 out of 4,753 hedge funds. Panel C contains the percentiles of the summary statistics of the mutual funds. The fund size is reported for 2,165 out of 3,905 mutual funds. Panel D contains the percentiles of additional hedge fund characteristics, i.e., management fee (mfee), performance fee (pfee), lockup period (lockup), advance notice period (adv notice), minimum investment amount (minv), and first available net asset value (first NAV). All hedge fund return statistics are based on monthly returns adjusted for smoothing and all mutual fund return statistics are based on simple returns. The sample period is, after deleting the first 12 months of returns for every hedge fund, from January 1994 to December 2014.

Panel A: benchmark assets and portfolios of hedge funds									
	T	mean (in %)	std (in %)	skew	kurt	JB	min (in %)	max (in %)	N funds
stocks	252	0.63	4.32	−0.70	4.15	0.00	−16.87	10.93	
bonds	252	0.28	2.03	0.06	4.12	0.01	−6.75	8.51	
directional debt	191	0.58	2.76	−2.26	21.34	0.00	−19.91	9.88	207
directional equity	252	0.51	3.14	−0.55	4.74	0.00	−12.11	10.03	1,546
event	239	0.56	2.29	−1.45	8.64	0.00	−11.44	6.80	247
fund of funds	252	0.30	2.15	−0.69	5.90	0.00	−9.17	6.41	1,375
global derivatives	252	0.49	3.08	0.34	4.52	0.00	−9.01	11.74	602
multistrategy	222	0.60	1.84	−1.50	9.80	0.00	−10.72	5.87	247
relative value	236	0.36	1.58	−3.13	21.90	0.00	−11.52	4.71	313
other	210	0.33	2.88	−0.87	9.62	0.00	−16.16	11.24	216
all strategies	252	0.38	1.97	−0.51	4.78	0.00	−8.45	5.80	4,753
Panel B: hedge funds									
	T	mean (in %)	std (in %)	skew	kurt	JB	min (in %)	max (in %)	fund size (\$ MM)
25th perc	57	0.12	2.19	−0.82	3.62	0.00	−17.64	5.63	16
median	87	0.37	3.26	−0.28	4.83	0.00	−10.55	9.29	55
75th perc	133	0.69	5.27	0.25	7.27	0.09	−6.36	16.99	173
mean	102	0.43	4.27	−0.32	7.18	0.10	−14.01	14.02	264
Panel C: mutual funds									
	T	mean (in %)	std (in %)	skew	kurt	JB	min (in %)	max (in %)	fund size (\$ MM)
25th perc	76	0.28	4.37	−0.72	3.76	0.00	−20.87	10.78	93
median	133	0.56	4.96	−0.57	4.30	0.00	−18.02	12.86	421
75th perc	203	0.75	5.72	−0.34	4.98	0.04	−15.35	16.22	1,507
mean	141	0.49	5.15	−0.53	4.51	0.07	−17.87	14.38	2,526
Panel D: additional characteristics of hedge funds									
		mfee (in %)	pfee (in %)	lockup (months)	adv notice (days)	minv (\$ MM)	first NAV (\$ MM)		
25th perc		1.00	10.00	0.00	30.00	0.10	2.27		
median		1.50	20.00	0.00	31.00	0.50	9.99		
75th perc		2.00	20.00	12.00	60.00	1.00	32.40		
mean		1.44	15.63	5.21	43.76	1.15	67.11		
reported (in %)		91.29	92.32	64.97	73.93	91.02	90.15		

Table 2: Intersection for portfolios of hedge funds

This table contains the results of the mean-variance-skewness intersection tests in which the test assets are net asset value weighted portfolios of hedge fund returns within one Morningstar category and the benchmark assets are the S&P 500 Total Return Index and 10 year US government bonds. The left-hand side of the table shows the results of the intersection tests for $\gamma_1 = 5$ and $\gamma_2 = 15$. The corresponding weights invested in the benchmark assets are 0.73 (stocks) and 1.55 (bonds) if a risk-free asset is available and 0.48 (stocks) and 0.52 (bonds) if no risk-free asset is available. The right-hand side of the table shows the results of the intersection tests for $\gamma_1 = \gamma_2 = 15$. The corresponding weights invested in the benchmark assets are 0.26 (stocks) and 0.54 (bonds) if a risk-free asset is available and 0.30 (stocks) and 0.70 (bonds) if no risk-free asset is available. The Wald statistics are calculated with White [1980] standard errors. P-values are reported below the Wald statistics in parenthesis and 5% significance is indicated in bold.

		[γ_1, γ_2] = [5.00, 15.00]				[γ_1, γ_2] = [15.00, 15.00]			
		risk-free asset				risk-free asset			
		with		no		with		no	
		short-sales		short-sales		short-sales		short-sales	
		with	no	with	no	with	no	with	no
directional debt	wstat	2.76	2.76	4.82	4.82	3.79	3.79	6.89	6.89
	pval	(0.097)	(0.048)	(0.028)	(0.014)	(0.051)	(0.026)	(0.009)	(0.004)
directional equity	wstat	2.31	2.31	1.85	1.85	2.44	2.44	6.90	6.90
	pval	(0.128)	(0.064)	(0.174)	(0.087)	(0.118)	(0.059)	(0.009)	(0.004)
event	wstat	7.40	7.40	4.55	4.55	8.25	8.25	14.65	14.65
	pval	(0.007)	(0.003)	(0.033)	(0.016)	(0.004)	(0.002)	(0.000)	(0.000)
fund of funds	wstat	0.51	0.51	0.14	0.14	0.59	0.59	3.89	3.89
	pval	(0.477)	(0.238)	(0.711)	(0.355)	(0.442)	(0.221)	(0.049)	(0.024)
global derivatives	wstat	4.10	4.10	3.42	3.42	3.90	3.90	7.56	7.56
	pval	(0.043)	(0.021)	(0.064)	(0.032)	(0.048)	(0.024)	(0.006)	(0.003)
multistrategy	wstat	18.58	18.58	9.47	9.47	19.91	19.91	27.54	27.54
	pval	(0.000)	(0.000)	(0.002)	(0.001)	(0.000)	(0.000)	(0.000)	(0.000)
relative value	wstat	7.76	7.76	1.35	1.35	9.13	9.13	15.53	15.53
	pval	(0.005)	(0.003)	(0.246)	(0.123)	(0.003)	(0.001)	(0.000)	(0.000)
other	wstat	0.24	0.24	0.21	0.21	0.38	0.38	1.71	1.71
	pval	(0.625)	(0.312)	(0.647)	(0.324)	(0.537)	(0.267)	(0.191)	(0.096)

Table 3: Spanning for portfolios of hedge funds

This table contains the results of the mean-variance (MV) and mean-variance-skewness (MVS) spanning tests in which the test assets are net asset value weighted portfolios of hedge fund returns within one Morningstar category and the benchmark assets are the S&P 500 TR Index and the 10 year US T-Bonds. The Wald statistics are calculated with White [1980] standard errors. P-values are reported below the Wald statistics in parenthesis and 5% significance is indicated in bold.

		with risk-free asset				no risk-free asset			
		with short-sales		no short-sales		with short-sales		no short-sales	
		MV	MVS	MV	MVS	MV	MVS	MV	MVS
directional debt	wstat	3.97	22.19	3.97	3.97	5.03	78.99	4.29	4.29
	pval	(0.046)	(0.000)	(0.023)	(0.174)	(0.081)	(0.000)	(0.056)	(0.191)

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directional equity	wstat	2.46	5.76	2.46	2.77	52.75	58.55	5.78	5.88
	pval	(0.117)	(0.217)	(0.058)	(0.262)	(0.000)	(0.000)	(0.026)	(0.097)
event	wstat	8.37	28.35	8.37	8.44	67.53	180.04	14.34	14.34
	pval	(0.004)	(0.000)	(0.002)	(0.023)	(0.000)	(0.000)	(0.000)	(0.002)
fund of funds	wstat	0.60	2.71	0.60	0.72	67.15	78.87	3.18	3.18
	pval	(0.437)	(0.607)	(0.219)	(0.620)	(0.000)	(0.000)	(0.097)	(0.292)
global derivatives	wstat	3.87	7.79	3.87	7.52	21.90	30.82	6.33	9.97
	pval	(0.049)	(0.100)	(0.025)	(0.032)	(0.000)	(0.000)	(0.019)	(0.017)
multistrategy	wstat	20.10	49.15	20.10	20.18	136.89	295.91	30.87	30.87
	pval	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
relative value	wstat	9.34	34.72	9.34	9.34	116.19	279.77	21.05	21.05
	pval	(0.002)	(0.000)	(0.001)	(0.016)	(0.000)	(0.000)	(0.000)	(0.000)
other	wstat	0.41	6.90	0.41	0.41	18.35	86.93	1.10	1.73
	pval	(0.524)	(0.141)	(0.261)	(0.679)	(0.000)	(0.000)	(0.289)	(0.508)

Table 4: Spanning for individual hedge funds

The table reports average rejection rates of the mean-variance and mean-variance-skewness spanning tests with stocks and bonds as benchmark assets and individual hedge funds as test assets. Rejection rates are reported in percent and at the 5% significance level. To analyze cross-sectional effects, hedge funds are sorted independently in quintiles according to the intercept of the mean-variance spanning regression and in quintiles according to the residual co-skewness Wald statistic. The horizontal quintiles correspond to the sort on alpha ("low alpha" to "high alpha") and the vertical quintiles to the sort on the residual co-skewness Wald statistic ("low coskew" to "high coskew"). The spanning test assumes that a risk-free asset is available and Wald statistics are calculated with White [1980] standard errors.

	low alpha	Q2	Q3	Q4	high alpha	all
Panel A: with short-sales						
MV						
low coskew	13.70	0.50	11.16	33.02	60.11	23.34
Q2	12.10	0.00	11.06	29.03	55.05	21.58
Q3	10.06	0.54	9.55	40.11	57.08	24.50
Q4	7.28	0.00	11.48	35.59	53.96	21.89
high coskew	10.26	0.55	8.90	31.91	53.42	19.77
all	10.41	0.32	10.52	33.89	55.94	22.22
MVS						
low coskew	6.16	0.50	6.51	14.62	30.90	11.57
Q2	7.64	0.50	11.54	19.89	39.39	16.00
Q3	10.65	2.72	15.58	41.71	49.06	24.82
Q4	22.33	15.93	40.44	61.02	68.81	41.68
high coskew	93.04	91.80	95.21	98.40	99.38	95.27
all	35.65	21.47	29.65	46.21	56.36	37.87
Panel B: no short-sales						
MV						
low coskew	0.00	0.00	22.79	49.53	73.60	29.97
Q2	0.00	0.00	17.31	40.32	69.19	26.11
Q3	0.00	1.09	19.10	51.34	69.34	29.76
Q4	0.00	0.00	18.58	49.15	67.82	27.16
high coskew	0.00	1.64	14.38	42.55	67.08	22.29
all	0.00	0.53	18.72	46.63	69.40	27.06
MVS						
low coskew	0.00	0.00	6.05	21.70	44.38	14.51
Q2	0.00	0.00	11.54	23.66	49.49	17.47
Q3	0.59	1.63	16.58	44.39	57.55	25.45
Q4	12.14	25.27	33.88	61.02	62.87	38.74
high coskew	51.28	50.82	63.01	84.57	78.88	64.25
all	17.46	14.95	23.55	46.32	58.15	32.08

Table 5: Hedge fund characteristics and spanning test statistics

The table analyzes the relation between the spanning test statistics and hedge fund characteristics. The dependent variable for Model (1) to (4) is the t-statistic of the intercept of the mean-variance regression and the dependent variable for Model (5) to (8) is the logarithm of 1 plus Wald statistic of the mean-variance-skewness spanning test with short-sales constraints calculated for Table 4. The independent variables are the management fee in percent (mfee), the performance fee in percent (pfee), a dummy equal to 1 if the fund has a high watermark provision (dhwater), the minimum investment amount in million of USD (minv), the logarithm of 1 plus the sum of the number of lockup months and the advanced notice days divided by 30 (loglocknot), the logarithm of the net asset value of the fund at inception (logsize), the standard deviation and the coefficient of skewness of the returns of the fund (std and skew), and dummies for all Morningstar categories except directional equity, i.e., for directional debt (ddebit), event (event), fund of funds (fof), global derivatives (gderivatives), multistrategy (mstrategy), relative value (rvalue), other (other). The dependent and independent variables (except dummies) are winsorized at the 1% percent level. The p-values are reported under the coefficients in parentheses and the last two rows of the table report the adjusted r^2 and the number of observations in the regression. Coefficients significant at the 5% percent level are highlighted with bold font.

dep. var.	t-stat alpha				wstat MVS span no short-sales			
model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
constant	0.669 (0.000)	0.905 (0.000)	1.869 (0.000)	2.117 (0.000)	1.196 (0.000)	1.551 (0.000)	1.771 (0.000)	1.920 (0.000)
mfee	-0.004 (0.941)		0.106 (0.071)	0.089 (0.132)	0.056 (0.076)		0.089 (0.021)	0.062 (0.118)
pfee	0.020 (0.000)		0.023 (0.000)	0.009 (0.174)	0.017 (0.000)		0.018 (0.000)	0.007 (0.110)
dhwater	0.048 (0.516)		-0.138 (0.134)	-0.074 (0.422)	0.082 (0.074)		-0.010 (0.872)	0.037 (0.551)
minv		0.093 (0.000)	0.066 (0.000)	0.060 (0.001)		0.035 (0.003)	0.033 (0.008)	0.029 (0.018)
loglocknot		0.052 (0.067)	0.065 (0.018)	0.038 (0.182)		0.019 (0.287)	0.032 (0.076)	0.040 (0.036)
logsize			-0.033 (0.025)	-0.036 (0.014)			-0.020 (0.038)	-0.018 (0.062)
std			-15.999 (0.000)	-15.503 (0.000)			-6.511 (0.000)	-6.407 (0.000)
skew			0.351 (0.000)	0.357 (0.000)			0.116 (0.000)	0.114 (0.000)
ddebit				0.673 (0.000)				0.253 (0.008)
event				0.482 (0.000)				0.096 (0.233)
fof				-0.189 (0.063)				-0.192 (0.005)
gderivatives				-0.123 (0.226)				0.103 (0.130)
mstrategy				0.444 (0.001)				0.171 (0.056)
rvalue				0.096 (0.377)				0.113 (0.118)
other				-0.186 (0.194)				-0.201 (0.035)

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dep. var.	t-stat alpha				wstat MVS span no short-sales			
model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
adj R2	0.009	0.010	0.168	0.187	0.021	0.003	0.077	0.087
N	3851	2743	2451	2451	3851	2743	2451	2451

Table 6: Spanning for individual mutual funds

The table reports rejection rates for the individual mutual funds as test assets. The table is constructed in the same way as Table 4.

	low α	Q2	Q3	Q4	high α	all
Panel A: with short-sales						
MV						
low coskew	32.04	7.83	3.68	0.56	30.00	13.32
Q2	32.31	13.66	3.38	1.73	31.36	16.01
Q3	32.88	5.63	5.39	0.70	24.70	13.83
Q4	34.25	6.47	7.19	0.00	20.00	13.06
high coskew	34.38	12.26	5.15	0.88	25.62	18.69
all	33.42	9.22	4.99	0.77	26.50	14.98
MVS						
low coskew	16.50	4.22	3.07	0.00	7.06	5.25
Q2	16.92	10.56	3.38	0.58	16.57	9.35
Q3	21.23	5.00	5.39	0.00	19.28	10.24
Q4	29.45	7.91	5.39	3.45	32.26	15.24
high coskew	75.39	56.77	46.32	57.52	74.38	63.89
all	39.18	16.77	11.65	9.22	27.14	20.79
Panel B: no short-sales						
MV						
low coskew	0.00	0.00	0.00	2.23	46.47	10.63
Q2	0.00	0.00	0.00	2.89	50.30	11.52
Q3	0.00	0.00	0.00	3.52	39.76	9.09
Q4	0.00	0.00	0.60	2.30	33.55	7.30
high coskew	0.00	0.00	0.00	2.65	43.80	7.17
all	0.00	0.00	0.13	2.69	42.89	9.14
MVS						
low coskew	0.00	0.00	0.00	0.00	12.35	2.69
Q2	0.00	0.00	0.00	0.58	13.61	3.07
Q3	0.00	0.00	0.00	1.41	12.05	2.82
Q4	0.68	1.44	1.20	0.57	13.55	3.46
high coskew	31.64	33.55	33.09	30.97	39.67	33.42
all	10.50	6.91	6.02	4.99	17.03	9.09

Table 7: Performance analysis for hedge funds

This table compares the alpha of individual hedge funds with respect to the benchmark assets (α) with the alpha adjusted for higher order co-moments up to order three (prudent alpha; α^{pr}), up to order four (temperant alpha; α^{te}) and all higher co-moments (α^{CRR}). To adjust alpha for higher order co-moments, we consider a portfolio of stocks and bonds which is chosen optimally on the respective sample for each hedge fund for a CRR investor with a coefficient of relative risk aversion of 10. The resulting average fractions invested in stocks and bonds are 0.48 and 1.09, respectively. The individual hedge funds are sorted independently in quintiles according to the intercept of the regression of the excess return of the fund on the excess return of the portfolio (horizontal quintiles: "low α " to "high α ") and the residual co-skewness of the hedge fund with the portfolio of benchmark assets (vertical quintiles: "low coskew" to "high coskew"). To maximize the dispersion in alphas and residual co-skewnesses, we divide each lowest and highest quintile in deciles. Panel A contains average alphas and Panel B contains $\alpha^{pr} - \alpha$. Panel C and Panel D report $\alpha^{te} - \alpha^{pr}$ and $\alpha^{CRR} - \alpha^{pr}$, respectively.

	low α	Q1.2	Q2	Q3	Q4	Q5.1	high α	all
Panel A: average annualized α in percent								
low coskew	-11.47	-2.14	0.64	2.82	5.49	8.75	20.27	1.75
Q1.2	-8.23	-2.15	0.63	2.91	5.57	8.67	17.57	3.70
Q2	-8.38	-2.04	0.75	2.89	5.56	8.82	15.73	3.30
Q3	-7.19	-1.93	0.68	2.90	5.53	8.79	16.86	3.65
Q4	-7.32	-2.08	0.68	2.94	5.52	8.81	15.77	3.37
Q5.1	-8.57	-2.10	0.56	2.90	5.71	9.22	17.38	4.05
high coskew	-12.13	-2.08	0.45	3.10	5.78	8.90	17.91	2.58
all	-9.70	-2.07	0.66	2.91	5.58	8.85	17.34	3.27
Panel B: average annualized $\alpha^{pr} - \alpha$ in percent								
low coskew	-1.43	-1.11	-0.85	-1.04	-1.19	-1.43	-1.52	-1.22
Q1.2	-0.28	-0.27	-0.27	-0.26	-0.26	-0.26	-0.26	-0.26
Q2	-0.07	-0.08	-0.07	-0.06	-0.07	-0.07	-0.08	-0.07
Q3	0.02	0.02	0.02	0.02	0.02	0.03	0.03	0.02
Q4	0.15	0.16	0.14	0.15	0.15	0.15	0.15	0.15
Q5.1	0.38	0.37	0.37	0.37	0.37	0.37	0.37	0.37
high coskew	1.46	1.05	1.08	1.08	1.01	1.00	1.27	1.18
all	0.02	-0.02	0.02	0.02	0.05	0.05	0.04	0.03
Panel C: average annualized $\alpha^{te} - \alpha^{pr}$ in percent								
low coskew	0.04	0.07	-0.09	-0.08	0.09	-0.06	0.09	0.01
Q1.2	-0.05	-0.01	0.02	-0.02	-0.02	-0.04	-0.05	-0.02
Q2	-0.02	-0.00	-0.01	-0.01	-0.01	-0.00	-0.01	-0.01
Q3	-0.02	-0.01	-0.00	-0.00	-0.00	-0.00	-0.00	-0.00
Q4	-0.08	-0.00	0.01	0.01	0.01	0.01	-0.01	0.00
Q5.1	-0.03	-0.04	0.02	0.02	0.03	0.05	0.04	0.02
high coskew	-0.19	-0.01	0.03	0.02	0.01	0.04	0.02	-0.03
all	-0.06	-0.00	-0.00	-0.00	0.01	0.00	0.01	-0.00
Panel D: average annualized $\alpha^{CRR} - \alpha^{pr}$ in percent								
low coskew	-0.01	0.01	-0.13	-0.13	0.02	-0.13	0.02	-0.05
Q1.2	-0.09	-0.04	-0.00	-0.04	-0.04	-0.09	-0.11	-0.05
Q2	-0.05	-0.02	-0.03	-0.02	-0.03	-0.02	-0.06	-0.03
Q3	-0.03	-0.02	-0.00	-0.01	-0.01	-0.01	-0.04	-0.01
Q4	-0.11	-0.00	0.01	0.02	0.01	0.02	-0.04	-0.00
Q5.1	-0.01	-0.02	0.03	0.04	0.04	0.07	0.05	0.03
high coskew	-0.15	0.02	0.05	0.09	0.05	0.08	0.11	0.02
all	-0.07	-0.01	-0.01	-0.01	0.00	-0.00	-0.01	-0.01

Table 8: Hedge fund characteristics and performance statistics

The table analyzes the relationship between simple alpha and residual co-skewness of Table 7 and hedge fund characteristics. The dependent variable for Model (1) to (4) is the simple alpha in percent and the dependent variable for Model (5) to (8) is residual co-skewness multiplied by 10^5 . The independent variables and the construction of the table are explained in the caption of Table 5. The dependent and independent variables (except dummies) are winsorized at the 1% percent level. Coefficients significant at the 5% level are in bold font.

dep. var.	alpha in percent				residual coskew times 10^5			
model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
constant	0.029 (0.391)	0.261 (0.000)	0.414 (0.000)	0.392 (0.000)	-0.008 (0.880)	0.118 (0.000)	-0.165 (0.308)	-0.174 (0.300)
mfee	0.037 (0.047)		0.050 (0.022)	0.037 (0.098)	-0.011 (0.703)		-0.008 (0.808)	-0.006 (0.859)
pfee	0.014 (0.000)		0.007 (0.000)	0.006 (0.007)	0.003 (0.192)		-0.002 (0.513)	-0.000 (0.973)
dhwat	-0.014 (0.595)		-0.011 (0.753)	-0.009 (0.795)	0.023 (0.603)		0.040 (0.466)	0.027 (0.630)
minv		0.024 (0.001)	0.029 (0.000)	0.025 (0.000)		0.000 (0.991)	0.012 (0.292)	0.014 (0.219)
loglocknot		0.016 (0.116)	0.016 (0.123)	0.016 (0.150)		-0.046 (0.004)	-0.044 (0.007)	-0.038 (0.025)
logsize			-0.018 (0.001)	-0.019 (0.001)			0.008 (0.339)	0.010 (0.261)
std			0.288 (0.404)	0.608 (0.092)			3.523 (0.000)	3.164 (0.000)
skew			0.150 (0.000)	0.151 (0.000)			0.009 (0.537)	0.008 (0.591)
ddebt				0.166 (0.002)				-0.369 (0.000)
event				0.153 (0.001)				-0.067 (0.355)
fof				0.032 (0.399)				0.007 (0.907)
gderivatives				0.070 (0.067)				0.009 (0.879)
mstrategy				0.154 (0.002)				-0.012 (0.878)
rvalue				0.024 (0.550)				0.006 (0.926)
other				0.062 (0.246)				-0.247 (0.004)
adj R2	0.032	0.005	0.140	0.146	0.000	0.002	0.017	0.026
N	3851	2743	2451	2451	3851	2743	2451	2451

Table 9: Performance analysis for mutual funds

This table compares the alpha of mutual funds with respect to the benchmark assets (α) with the alpha adjusted for higher order co-moments up to order three (α^{pr}), up to order four (α^{te}) and all higher co-moments (α^{CRRRA}). The table is constructed in the same way as Table 7. The average weights invested in stocks and bonds are 0.42 and 0.88, respectively.

	low α	Q1.2	Q2	Q3	Q4	Q5.1	high α	all
Panel A: average annualized α in percent								
low coskew	-9.62	-2.89	-1.24	0.24	1.68	3.38	7.24	-0.60
Q1.2	-7.24	-3.03	-1.21	0.25	1.74	3.36	6.09	1.46
Q2	-6.56	-2.91	-1.24	0.14	1.77	3.29	6.06	0.77
Q3	-6.45	-2.83	-1.04	0.09	1.60	3.33	6.31	0.12
Q4	-7.04	-2.78	-1.15	0.19	1.58	3.21	6.32	-0.05
Q5.1	-7.68	-2.86	-1.35	0.13	1.66	3.34	7.27	-0.49
high coskew	-8.22	-2.95	-1.26	0.13	1.62	3.31	7.04	-1.63
all	-7.80	-2.87	-1.17	0.15	1.67	3.32	6.47	0.04
Panel B: average annualized difference of $\alpha^{pr} - \alpha$ in percent								
low coskew	-0.62	-0.43	-0.37	-0.33	-0.39	-0.28	-0.30	-0.41
Q1.2	-0.16	-0.16	-0.17	-0.16	-0.16	-0.16	-0.16	-0.16
Q2	-0.07	-0.07	-0.08	-0.07	-0.08	-0.08	-0.08	-0.08
Q3	-0.01	-0.02	-0.01	-0.01	-0.02	-0.02	-0.02	-0.01
Q4	0.03	0.03	0.03	0.03	0.03	0.04	0.03	0.03
Q5.1	0.12	0.11	0.10	0.11	0.10	0.11	0.11	0.11
high coskew	0.50	0.50	0.42	0.39	0.34	0.34	0.53	0.44
all	-0.01	0.03	0.01	-0.00	-0.03	-0.06	-0.04	-0.01
Panel C: average annualized difference of $\alpha^{te} - \alpha^{pr}$ in percent								
low coskew	-0.08	-0.01	-0.04	-0.05	-0.01	-0.03	-0.04	-0.04
Q1.2	-0.04	-0.02	-0.03	-0.02	-0.04	-0.03	-0.03	-0.03
Q2	0.01	-0.01	-0.02	-0.02	-0.02	-0.03	-0.02	-0.02
Q3	-0.00	0.00	-0.01	-0.01	-0.02	-0.02	-0.01	-0.01
Q4	0.01	-0.00	-0.00	-0.00	-0.01	-0.01	-0.00	-0.00
Q5.1	0.00	-0.01	-0.00	0.01	-0.00	-0.01	-0.03	-0.00
high coskew	-0.04	-0.02	0.03	0.02	0.03	0.05	0.09	0.01
all	-0.03	-0.01	-0.01	-0.01	-0.01	-0.02	-0.01	-0.01
Panel D: average annualized difference of $\alpha^{CRRRA} - \alpha^{pr}$ in percent								
low coskew	-0.13	-0.00	-0.05	-0.05	-0.02	-0.05	-0.10	-0.06
Q1.2	-0.06	-0.01	-0.03	-0.02	-0.04	-0.05	-0.07	-0.05
Q2	-0.01	-0.01	-0.02	-0.02	-0.03	-0.05	-0.06	-0.03
Q3	0.00	0.00	-0.01	-0.01	-0.03	-0.03	-0.03	-0.01
Q4	0.02	0.00	0.00	-0.01	-0.01	-0.02	-0.01	-0.00
Q5.1	0.03	-0.00	0.01	0.02	-0.00	-0.02	-0.05	0.00
high coskew	-0.01	-0.02	0.05	0.04	0.06	0.07	0.10	0.03
all	-0.03	-0.00	-0.00	-0.01	-0.02	-0.03	-0.04	-0.02

Table 10: Spanning for different fund types

This table contains the rejection rates of spanning for various subsamples. Panel A reports the rejection rates separately for funds which a first net asset value available below and above the median, Panel B examines rejection rates separately for young funds (36 to 89 return observations) and old funds (at least 90 return observations), Panel C examines rejection rates for the earlier subsample until June 2004 and the more recent subsample, Panel D reports rejection rates for each Morningstar category, Panel E reports rejection rates for simple returns, i.e., not adjusted for return smoothing, and the smoothing adjusted returns when the first 24 observation are deleted for each fund instead of the first 12 observation as in the main line analysis, and Panel F examines the effect of including a delisting return of -25% and -50% for liquidated funds. All rejection rates are at the 5% significance level and the Wald statistics use White [1980] standard errors.

with short-sales		no short-sales		with short-sales		no shortsales	
MV	MVS	MV	MVS	MV	MVS	MV	MVS
Panel A: size							
first NAV below median				first NAV above or equal to median			
21.28	37.47	25.8	31.22	22.92	37.72	28.24	32.49
Panel B: age							
young				old			
18.41	39.67	20.12	33.39	26.27	35.95	34.39	30.7
Panel C: subperiod							
Jan 1994 to Jun 2004				Jul 2004 to Dec 2014			
39.15	42.45	47.07	40.66	16.08	39.58	18.62	33.49
Panel D: Morningstar categories							
directional debt				directional equity			
38.16	53.14	40.1	39.61	21.54	35.19	27.3	32.73
event				fund of funds			
40.49	48.99	48.18	42.91	13.16	30.47	16.58	22.62
global derivatives				multistrategy			
20.27	37.38	27.24	37.21	36.44	53.44	38.87	41.3
relative value				other			
28.12	51.12	32.59	39.94	29.17	41.2	32.41	31.48
Panel E: simple returns and deleting first 24 returns							
simple returns				delete first 24 returns			
31.96	47.91	36.21	34.95	21.03	35.92	24.53	30.05
Panel F: delisting return for liquidated funds							
delisting return of -25%				delisting return of -50%			
13.11	33.31	15.02	24.68	11.05	32.06	12.14	22.85

Table 11: Performance analysis for different fund types

This tables examines the robustness of the performance analysis results in Table 7. Panel A splits the sample in half as a function of the first net asset value available for each fund, Panel B examines alphas separately for young funds (36 to 89 return observations) and old funds (at least 90 return observations), Panel C examines alphas for the earlier subsample until June 2004 and the more recent subsample, Panel D reports alphas for each Morningstar category, Panel E reports alphas for simple returns, i.e., returns not adjusted for return smoothing, and alphas when the first 24 observation are deleted for each fund instead of the first 12 observation as in the base line analysis, and Panel F examines the effect of including a delisting return of -25% and -50% for liquidated funds. Panel G calculates alphas for a CRRA investor with relative risk aversion of 4 and a habit formation type investor with $u(x) = (x - k)^{1-\gamma}/(1 - \gamma)$ where $\gamma = 2$ and $k = 0.9430$ which corresponds to the γ and average habit level in the calibration of Campbell and Cochrane [1999]. The associated average portfolio weights are 1.19 for stocks and 2.69 for bonds in the former case and 0.13 for stocks and 0.30 for bonds in the latter case. Finally, Panel F forms quintiles based on $\alpha^{te} - \alpha^{pr}$ and $\alpha^{CRRA} - \alpha^{pr}$, respectively. The table reports the averages in the lowest two deciles, i.e., Q1.1 and Q1.2, the two highest deciles, i.e., Q5.1 and Q5.2, and the overall average.

	Q1.1	Q1.2	Q5.1	Q5.2	all	Q1.1	Q1.2	Q5.1	Q5.2	all
Panel A: size										
	first NAV below median					first NAV weakly above median				
α	2.72%	4.49%	4.56%	4.14%	4.02%	0.79%	3.49%	3.73%	0.61%	2.57%
$\alpha^{pr} - \alpha$	-1.26%	-0.27%	0.40%	1.29%	0.04%	-1.04%	-0.22%	0.33%	1.02%	0.03%
Panel B: age										
	young					old				
α	0.95%	2.21%	1.98%	1.64%	2.26%	4.55%	4.04%	5.09%	5.60%	4.35%
$\alpha^{pr} - \alpha$	-1.76%	-0.50%	0.51%	1.57%	0.01%	-0.47%	-0.14%	0.26%	0.67%	0.05%
Panel C: subperiod										
	Jan 1994 to Jun 2004					Jul 2004 to Dec 2014				
α	8.17%	9.14%	8.60%	11.74%	8.03%	1.16%	2.58%	2.76%	2.16%	2.37%
$\alpha^{pr} - \alpha$	-0.35%	-0.14%	0.08%	0.29%	-0.02%	-1.31%	-0.28%	0.47%	1.30%	0.07%
Panel D: Morningstar categories										
	directional debt					directional equity				
α	7.44%	2.54%	3.27%	5.33%	4.06%	0.80%	5.41%	2.44%	1.16%	3.68%
$\alpha^{pr} - \alpha$	-1.79%	-0.69%	0.28%	1.01%	-0.16%	-1.34%	-0.29%	0.53%	1.48%	0.08%
	event					fund of funds				
α	6.57%	7.29%	4.76%	3.96%	5.54%	0.14%	1.03%	1.18%	0.44%	1.28%
$\alpha^{pr} - \alpha$	-1.13%	-0.36%	0.28%	0.90%	-0.04%	-0.70%	-0.14%	0.24%	0.64%	0.02%
	global derivatives					multistrategy				
α	3.06%	4.50%	7.94%	5.82%	5.17%	-2.18%	4.03%	6.35%	3.58%	4.09%
$\alpha^{pr} - \alpha$	-1.78%	-0.29%	0.52%	1.72%	0.07%	-1.43%	-0.50%	0.36%	1.24%	-0.03%
	relative value					other				
α	2.75%	3.77%	4.71%	3.28%	3.38%	-1.96%	0.91%	2.30%	4.10%	3.32%
continued on the next page										

$\alpha^{pr} - \alpha$	-0.88%	-0.22%	0.29%	0.87%	0.00%	-1.11%	-0.41%	0.11%	0.68%	-0.10%
Panel E: simple returns and deleting first 24 returns										
α	simple returns					delete first 24 returns				
	2.02%	3.88%	4.55%	3.84%	3.70%	0.52%	3.05%	3.17%	1.24%	2.74%
	$\alpha^{pr} - \alpha$	-1.25%	-0.33%	0.24%	0.95%	-0.05%	-1.24%	-0.26%	0.40%	1.41%
Panel F: delisting return for liquidated funds										
α	delisting return of -25%					delisting return of -50%				
	-0.73%	0.71%	0.97%	-1.14%	0.64%	-4.60%	-1.18%	-2.80%	-5.82%	-1.98%
	$\alpha^{pr} - \alpha$	-1.36%	-0.31%	0.43%	1.35%	0.04%	-1.71%	-0.35%	0.51%	1.66%
Panel G: alternative utility										
α	CRRA with RRA of 4					habit				
	1.81%	3.60%	4.22%	2.39%	3.27%	1.90%	3.75%	4.23%	2.36%	3.27%
	$\alpha^{pr} - \alpha$	-1.25%	-0.28%	0.40%	1.25%	0.03%	-1.24%	-0.30%	0.43%	1.31%
	$\alpha^{te} - \alpha^{pr}$	0.03%	-0.02%	0.02%	-0.06%	-0.01%	0.05%	-0.02%	0.02%	-0.14%
	$\alpha^{CRRA} - \alpha^{pr}$	-0.16%	-0.07%	0.05%	0.08%	-0.01%	-0.49%	-0.18%	0.15%	0.27%
Panel H: alternative sorting										
α	sorting on $\alpha^{te} - \alpha^{pr}$					sorting on $\alpha^{CRRA} - \alpha$				
	3.55%	3.92%	2.59%	1.11%	3.27%	1.98%	3.74%	3.63%	3.07%	3.27%
	$\alpha^{pr} - \alpha$	0.01%	0.09%	0.02%	0.08%	0.03%	-1.13%	-0.27%	0.37%	1.11%
	$\alpha^{te} - \alpha^{pr}$	0.38%	0.07%	-0.08%	-0.40%	0.00%	-0.12%	0.00%	0.02%	0.08%
	$\alpha^{CRRA} - \alpha^{pr}$	0.39%	0.08%	-0.10%	-0.43%	-0.01%	-0.20%	-0.06%	0.04%	0.14%

Table 12: Simulated option strategies

This table reports alphas, betas and higher-order residual co-moments of the strategy long the market and short one or five call options in Panel A and long the market and long one or five put options in Panel B. The market follows a logarithmic Brownian motion with annual expected return of 12% and annual volatility of 15%. The initial value of the market is 100, the horizon is one year and the risk-free rate equals 5%. The statistics in the table are calculated using 10 million simulated terminal values of the market and option prices are calculated with the Black-Scholes model. The example with one option is from Leland [1999]. The table reports simple alphas (α), prudent alphas (α^{pr}), temperant alphas (α^{te}) and infinity/all moments ($\alpha^{CRR A}$). For the approximations, we use the gamma of 3.63, which is obtained by choosing the gamma such that a 100% investment in the market is optimal. We also report the beta of the strategy (β), higher order residual co-moments (coskew and cokurt), and the beta with respect to the CRR A utility function ($\beta^{CRR A}$).

strike	α	α^{pr}	α^{te}	$\alpha^{CRR A}$	β	coskew	cokurt	$\beta^{CRR A}$
Panel A: long the market and short n call option(s)								
$n = 1$								
110	0.87	-0.04	0.12	-0.00	0.41	-0.14	-0.01	0.53
120	0.73	-0.16	0.14	-0.00	0.66	-0.13	-0.03	0.76
130	0.43	-0.18	0.14	-0.00	0.84	-0.09	-0.03	0.90
$n = 5$								
110	5.07	-0.22	0.68	-0.01	-2.44	-0.79	-0.08	-1.72
120	3.84	-0.85	0.74	-0.01	-0.80	-0.70	-0.14	-0.25
130	2.21	-0.90	0.68	-0.00	0.19	-0.46	-0.14	0.51
Panel B: long the market and long n put option(s)								
$n = 1$								
90	-0.24	-0.07	-0.04	0.00	0.95	0.03	-0.00	0.92
100	-0.60	-0.08	-0.08	0.00	0.81	0.08	0.00	0.73
110	-0.78	0.03	-0.10	0.00	0.56	0.12	0.01	0.45
$n = 5$								
90	-1.17	-0.35	-0.21	0.00	0.78	0.12	-0.01	0.61
100	-2.66	-0.35	-0.35	0.01	0.16	0.35	0.00	-0.22
110	-2.99	0.13	-0.40	0.01	-0.69	0.47	0.05	-1.12

Table 13: Alternative factor models for the simulated option strategies

This table contains coefficients of the regression of the strategies in Table 12 on the excess market return and the squared excess market return (quadratic model), on the excess market return and the return on the call option strategy (market + call), on the excess market return and the excess return on the put option strategy (market + put), and on the excess return of the call and put option strategy (call + put). The call (put) option strategy consists in buying call (put) option with strike 100 at the Black-Scholes price and to hold the option until maturity. The intercept of each regression is reported in percent.

strike	quadratic model			market + call			market + put			call + put		
	alpha	rm	rm^2	alpha	rm	option	alpha	rm	option	alpha	call	put
Panel A: long the market and short n call option(s)												
$n = 1$												
110	2.56	0.66	-1.26	10.27	1.43	-0.10	4.16	0.21	-0.04	3.12	0.02	-0.05
120	2.39	0.91	-1.24	7.36	1.38	-0.07	3.05	0.52	-0.03	0.47	0.04	-0.04
130	1.57	1.01	-0.85	3.87	1.21	-0.04	1.64	0.77	-0.01	-2.20	0.06	-0.04
$n = 5$												
110	14.87	-0.97	-7.33	59.73	3.51	-0.56	24.20	-3.60	-0.22	42.20	-0.28	-0.11
120	12.54	0.50	-6.51	38.66	2.99	-0.36	16.02	-1.54	-0.14	23.71	-0.12	-0.09
130	7.96	1.05	-4.30	19.63	2.09	-0.18	8.30	-0.18	-0.07	9.20	-0.01	-0.07
Panel B: long the market and long n put option(s)												
$n = 1$												
90	-0.56	0.91	0.24	-2.89	0.67	0.03	-1.17	1.01	0.01	-6.22	0.08	-0.02
100	-1.56	0.67	0.72	-8.05	0.00	0.08	-3.21	0.97	0.03	-8.05	0.08	-0.00
110	-2.28	0.33	1.13	-9.17	-0.35	0.09	-3.72	0.74	0.03	-7.41	0.06	0.01
$n = 5$												
90	-2.71	0.55	1.15	-14.05	-0.62	0.13	-5.68	1.05	0.05	-10.94	0.08	0.02
100	-6.96	-0.48	3.21	-35.89	-3.46	0.34	-14.29	0.86	0.14	-18.61	0.07	0.11
110	-8.78	-1.56	4.33	-35.25	-4.20	0.33	-14.28	-0.01	0.13	-14.25	-0.00	0.13

Internet Appendix

Table 1: Additional simulated option strategies

Panel A and B contain the performance statistics of buy and hold strategies of Call options and straddles, respectively. The strikes of these options vary from 90 to 120. The table reports simple alphas (α) and prudent alphas (i.e., alphas adjusted for higher order co-moments up to order three; α^{pr}) and infinity/all moments (α^{CRR}). The table also contains the coefficients of single factor models with excess returns on straddles with strike 100 (Str_{100}) and call options with strike 100 ($Call_{100}$) and of a two factor model with excess returns on call options with strike 100 and strike 120 ($Call_{120}$) as factors. These strategies are calculated in the same way and with the same parameters than in Table 12 in the paper.

strike	α	α^{pr}	α^{CRR}	alpha	Str_{100}	alpha	$Call_{100}$	alpha	$Call_{100}$	$Call_{120}$
Panel A: call option										
90	-1.63%	-0.49%	-0.01%	18.07%	0.80	4.71%	0.58	3.56%	0.65	-0.02
95	-3.81%	-0.83%	-0.01%	19.95%	1.06	3.40%	0.76	2.38%	0.81	-0.02
100	-7.83%	-1.05%	-0.01%	20.56%	1.45	0.00%	1.00	0.00%	1.00	0.00
105	-14.54%	-0.70%	-0.01%	19.63%	1.98	-6.10%	1.33	-2.75%	1.13	0.07
110	-24.94%	0.99%	-0.01%	16.59%	2.66	-15.79%	1.75	-4.61%	1.09	0.22
115	-40.13%	5.15%	-0.03%	10.58%	3.53	-30.19%	2.29	-3.97%	0.75	0.52
120	-61.34%	13.38%	-0.09%	0.68%	4.62	-50.51%	2.97	0.00%	0.00	1.00
Panel B: straddle										
90	-3.11%	-0.94%	-0.02%	13.82%	0.76	2.00%	0.54	1.15%	0.59	-0.02
95	-6.63%	-1.45%	-0.02%	9.41%	0.93	-2.66%	0.62	-2.78%	0.63	0.00
100	-11.20%	-1.51%	-0.02%	0.00%	1.00	-9.96%	0.63	-8.05%	0.51	0.04
105	-14.54%	-0.70%	-0.01%	-10.98%	0.86	-16.47%	0.49	-11.13%	0.18	0.11
110	-14.65%	0.58%	-0.01%	-18.45%	0.53	-18.82%	0.26	-10.42%	-0.24	0.17
115	-12.08%	1.55%	-0.01%	-20.69%	0.19	-17.17%	0.04	-7.73%	-0.52	0.19
120	-8.71%	1.90%	-0.01%	-19.44%	-0.06	-13.79%	-0.11	-5.35%	-0.61	0.17